

Entropy mixing and stability under Boolean noise

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Abstract

We present an analytic proof candidate for the Courtade–Kumar inequality, together with a dimension-independent stability estimate. The argument is organized around three statements that can be used separately. A restriction–mixing criterion converts a two-posterior entropy inequality into a bound in every dimension. An entropy-product comparison identifies the symmetric pair as an extremizer for the product of mixing gain and post-mixing entropy. A spectral-cap certificate converts a scalar entropy envelope into an information bound. At low noise, the first two statements control all output means by induction. At complementary noise, an antipodal identity and a polynomial moment bound supply the spectral cap. Retaining the mixing surplus gives a second induction and an explicit bound on distance to the nearest coordinate. Complete scalar proofs, exact finite-constant bounds, and examples are included. The paper uses no specialized external theorem as a premise.

1 The problem, the results, and the mechanisms

Let $X = (X_1, \dots, X_n)$ be uniform on $\{-1, 1\}^n$, and obtain Y by independently reversing each sign with probability $p \in [0, 1/2]$; put $\rho = 1 - 2p$. A deterministic Boolean function $f : \{-1, 1\}^n \rightarrow \{0, 1\}$ retains one bit of X . The question is whether combining coordinates can retain more information about Y than simply keeping one coordinate.

All logarithms are natural. Write

$$h(s) = -s \ln s - (1 - s) \ln(1 - s), \quad 0 \ln 0 := 0,$$

and let $\mu = \mathbb{E}f(X)$. Then

$$I(f(X); Y) = h(\mu) - H(f(X) | Y). \quad (1.1)$$

For $f(x) = \mathbf{1}\{x_i = 1\}$, the right side is $\ln 2 - h(p)$. Courtade and Kumar conjectured that this is the optimum over all Boolean functions [1].

Theorem 1.1 (Courtade–Kumar inequality). *For every finite n , every Boolean f , and every $p \in [0, 1/2]$,*

$$I(f(X); Y) \leq \ln 2 - h(p). \quad (1.2)$$

The statement uses uniform independent input bits and independent binary symmetric noise. It does not concern a general input distribution or a general channel. Expressing every entropy in bits divides both sides by $\ln 2$.

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1.1 Quantitative content

The proof gives more than an extremal value. At low noise it establishes the mean-dependent profile

$$H(f(X) | Y) \geq V(\mu)h(p), \quad V(\mu) = 4\mu(1 - \mu). \quad (1.3)$$

This holds for $0 \leq p \leq 1/20$. The factor $V(\mu)$ is four times the output variance. It is the quantity that makes restriction into smaller Boolean functions compatible with induction.

For $n \geq 1$, define the distance to a coordinate, allowing complementation, by

$$\delta(f) = \min_{i, \sigma \in \{-1, 1\}} \mathbb{P}\{2f(X) - 1 \neq \sigma X_i\}. \quad (1.4)$$

The quantitative conclusion is

$$\ln 2 - h(p) - I(f(X); Y) \geq \frac{p(1 - 2p)^2}{2000} \delta(f), \quad 0 \leq p \leq \frac{1}{2}. \quad (1.5)$$

Thus, at a fixed $0 < p < 1/2$, near-optimal information forces closeness to a coordinate, independently of n . The explicit constants are sufficient, not sharp. Section 7 gives a stronger piecewise coefficient, a matching qualitative converse, and worked examples. The stability results are consequences of the argument, not premises of Theorem 1.1.

1.2 Three ideas that can be used separately

Restriction needs a compensating quantity. If the two coordinate sections have means μ_0, μ_1 and $d = |\mu_1 - \mu_0|$, then

$$\frac{V(\mu_0) + V(\mu_1)}{2} = V(\mu) - d^2.$$

A local entropy inequality must pay for this precise loss. Proposition 2.3 formulates the resulting scalar-to-cube implication with a free profile coefficient c ; it is not tied to the numerical cutoff used in the application.

The useful symmetric quantity is a product. For two posterior probabilities, let J be the average entropy before mixing, M the average entropy after mixing, and $K = M - J$ the gain. The scalar problem has the form

$$K + Md^2 - 2\ell d = M \left(d - \frac{\ell}{M} \right)^2 + \frac{KM - \ell^2}{M}.$$

This is why the product KM is compared with its symmetric value. Lemma 3.1 proves that both K and KM are smallest at symmetry for a fixed separation. Its noise range is the entire open interval $0 < p < 1/2$, although the stronger two-point estimate needed for CK is used only at low noise.

A scalar envelope turns spectral information into entropy. Away from low noise, a bound on constant and linear Fourier mass controls a quadratic statistic of the posterior. The quadratic is dictated by a spectral cancellation, not guessed. Proposition 5.2 states the resulting information certificate with a free Fourier cap and a free entropy envelope. It separates what is special to the application from what can be reused.

1.3 Proof organization

Sections 2–3 establish the low-noise branch. Sections 4–5 establish the complementary branch, and Section 5.7 joins them at $p = 1/20$. Section 6 quantifies the slack. Section 7 explains how to apply the conclusions and what their constants do not say. Appendices A–E contain every deferred scalar and finite-constant argument.

For the main theorem, the task is therefore finite and explicit: prove one all-posterior mixing statement at low noise, and one Fourier-to-entropy estimate for the complementary class. No finite truth-table experiment is used to infer a statement in arbitrary dimension.

1.4 Relation to earlier work

Samorodnitsky proved the Courtade–Kumar inequality in a dimension-independent high-noise interval [2]. Yu’s extended paper proves local optimality among balanced functions and a computer-assisted balanced result for $0 \leq \rho \leq 0.914$ [3]. Javanmard and Woodruff prove an arbitrary-bias coordinate-wise information inequality and sharpen high-noise entropy estimates [4]. Their coordinate-wise sum is a different quantity from information about the full noisy vector.

The present proposed argument addresses all output means, all finite dimensions, and all noise levels, with the explicit stability estimate above. The ordinary variance identity, antipodal projection, and uniform-reference contraction are not claimed as new. First-level stability is classical [7]. The mathematical content to examine here is the compatibility of the local mixing inequality with restriction, the entropy-product comparison, and their combination with the Fourier certificate. None of the cited specialized results is a premise of the proof.

1.5 Notation and normalization

$$\rho = 1 - 2p, \quad \eta(u) = h\left(\frac{1+u}{2}\right), \quad \kappa(u) = \ln 2 - \eta(u), \quad -1 \leq u \leq 1. \quad (1.6)$$

Thus $\eta(\rho) = h(p)$ and $\kappa(\rho) = \ln 2 - h(p)$. The indicator f and the sign function $F = 2f - 1$ encode the same output bit; write $m = \mathbb{E}F = 2\mu - 1$. A signed coordinate means $F(x) = \sigma x_i$, or equivalently $f(x) = \mathbf{1}\{\sigma x_i = 1\}$. Fourier weights are introduced in Section 4.3.

On $(-1, 1)$,

$$\eta'(u) = -\operatorname{atanh} u, \quad \eta''(u) = -\frac{1}{1-u^2}.$$

Integrating the power series for atanh gives

$$\kappa(u) = \sum_{j \geq 1} \frac{u^{2j}}{(2j)(2j-1)}. \quad (1.7)$$

The coefficients are positive, and hence

$$\kappa(u) \geq \frac{u^2}{2}, \quad \kappa(\theta u) \leq \theta^2 \kappa(u) \quad (0 \leq \theta \leq 1). \quad (1.8)$$

Endpoint values follow by continuity. The numerical comparisons used below are proved by rational enclosures in Appendix E.

Two parameters must remain distinct: d is the fixed difference of section means; $t = |a - b|$ is the separation of a posterior pair. In an application $|A - B|$ varies with the noisy observation while d remains fixed. No independence between A and B is required.

2 From posterior mixing to a dimension-independent bound

A restriction of a balanced function need not be balanced. We therefore work with the whole mean-dependent profile, not only its value at $\mu = 1/2$. This section isolates the probability identities and the induction. The entropy estimate that supplies its hypothesis is proved in Section 3.

2.1 Restriction and the exact variance loss

Fix an input coordinate and let f_0, f_1 be the restrictions obtained by setting that coordinate to $-1, +1$. Let their means be μ_0, μ_1 . The parent mean and section-mean gap are

$$\mu = \frac{\mu_0 + \mu_1}{2}, \quad d = |\mu_1 - \mu_0|.$$

The law of total variance, or direct expansion, gives

$$\frac{V(\mu_0) + V(\mu_1)}{2} = V(\mu) - d^2. \quad (2.1)$$

Indeed, the variance of the two equally likely numbers μ_0, μ_1 is $d^2/4$. Multiplying the total-variance identity by four yields (2.1).

This identity determines what the induction needs. The two child bounds provide only $[V(\mu) - d^2]h(p)$; the desired parent bound is $V(\mu)h(p)$. We must recover the missing contribution when the noisy coordinate recombines the children.

For example, the two-bit AND function has one constant section and one coordinate section. Their means are 0 and $1/2$, whereas the parent mean is $1/4$. Thus $d = 1/2$, $V(\mu) = 3/4$, and the average child factor is $1/2 = 3/4 - (1/2)^2$. An induction confined to balanced outputs would not even admit this example.

2.2 What noisy recombination looks like

Suppose that, after seeing the other noisy coordinates, the probabilities that the two sections output 1 are a and b . The remaining noisy coordinate mixes them into

$$(1 - p)a + pb \quad \text{or} \quad pa + (1 - p)b,$$

with equal probabilities. Define the average resulting entropy by

$$M_p(a, b) = \frac{h((1 - p)a + pb) + h(pa + (1 - p)b)}{2}. \quad (2.2)$$

Concavity alone says that this is at least $[h(a) + h(b)]/2$. That only preserves the child entropy; it does not recover the variance loss. After recording the coordinate and channel identities, we state a criterion that isolates exactly what an induction needs. Its substantive hypothesis is scalar and does not refer to a dimension.

2.3 Selecting and observing a coordinate

A coordinate with d close to one would demand a stronger scalar statement. We need not choose that coordinate.

Lemma 2.1 (A small-gap coordinate). *If $n \geq 2$, at least one coordinate has section-mean gap at most $1/2$.*

Proof. Write $F = 2f - 1$ and $b_i = \mathbb{E}(FX_i)$. Conditioning on X_i shows that

$$b_i = \mathbb{E}f_1 - \mathbb{E}f_0.$$

For any distinct i, j , choose signs σ_i, σ_j to match the signs of b_i, b_j . Since $|F| = 1$,

$$|b_i| + |b_j| = \mathbb{E}[F(\sigma_i X_i + \sigma_j X_j)] \leq \mathbb{E}|\sigma_i X_i + \sigma_j X_j| = 1.$$

The last absolute value is zero or two, each with probability $1/2$. Therefore two coordinate gaps cannot both exceed $1/2$. When $n \geq 2$, some gap is at most $1/2$. $\square$

Notice the direction of the choice: the induction selects a small-gap coordinate, rather than requiring the scalar lemma to work for the most influential coordinate. One-bit functions are treated separately.

The posterior calculation. The induction must use the original probability model, rather than an arbitrary pair of random probabilities. We record the identity that makes the scalar lemma applicable.

Lemma 2.2 (Posterior mixtures). *Let X' be uniform on $\{-1, 1\}^{n-1}$, let U be an independent uniform sign, and let W, S be their respective independent BSC observations with flip probability p . Let f_0, f_1 be the sections of $f(X', U)$ at $U = -1, +1$. Define*

$$A(w) = \mathbb{P}\{f_0(X') = 1 \mid W = w\}, \quad B(w) = \mathbb{P}\{f_1(X') = 1 \mid W = w\}.$$

Then S is independent of W and fair, and

$$\begin{array}{c|c} \text{observation} & \mathbb{P}\{f(X', U) = 1 \mid W = w, S\} \\ \hline S = -1 & (1-p)A(w) + pB(w) \\ S = +1 & pA(w) + (1-p)B(w). \end{array} \quad (2.3)$$

Consequently $\mathbb{H}(f(X', U) \mid W, S) = \mathbb{E}M_p(A, B)$.

Proof. For k coordinates write the channel matrix as

$$\mathcal{K}_{p,k}(w, x) = \prod_{j=1}^k [(1-p)\mathbf{1}_{\{w_j=x_j\}} + p\mathbf{1}_{\{w_j \neq x_j\}}].$$

It is symmetric and every row sums to one. With uniform input, the output is uniform and Bayes' formula gives

$$\mathbb{P}\{X' = x \mid W = w\} = \mathcal{K}_{p,n-1}(w, x).$$

Thus $A(w) = \sum_x f_0(x) \mathcal{K}_{p,n-1}(w, x)$, and similarly for B . Independence of the two channel blocks implies that, given (W, S) , the posterior of X' depends only on W and that of U only on S . The latter assigns probabilities $1-p, p$ to the agreeing and disagreeing signs. Summing over U gives (2.3). Averaging its two binary entropies proves the last assertion. This uses no independence between $A(W)$ and $B(W)$. $\square$

2.4 The scalar-to-cube principle

Proposition 2.3 (Restriction–mixing criterion). *Fix $0 < p < 1/2$ and $0 \leq c \leq h(p)$. Suppose that, for every $a, b \in [0, 1]$ and independently every $d \in [0, 1/2]$,*

$$(1 + d^2)M_p(a, b) \geq \frac{h(a) + h(b)}{2} + 2cd|a - b|. \quad (2.4)$$

Then, in every finite dimension and at every output mean,

$$H(f(X) | Y) \geq cV(\mu). \quad (2.5)$$

Proof. Induct on dimension. A zero-bit function is constant. A one-bit function is constant or a signed coordinate; the latter has conditional entropy $h(p) \geq c$ and $V(\mu) = 1$.

For $n \geq 2$, select a coordinate with $d = |\mu_1 - \mu_0| \leq 1/2$ by Lemma 2.1. Use the posteriors A, B of Lemma 2.2. Its identity and the tower property give

$$H(f(X) | Y) = \mathbb{E}M_p(A, B), \quad \mathbb{E}|A - B| \geq |\mathbb{E}A - \mathbb{E}B| = d. \quad (2.6)$$

The induction hypothesis applies to both sections with their actual means, so

$$\mathbb{E} \frac{h(A) + h(B)}{2} \geq \frac{cV(\mu_0) + cV(\mu_1)}{2} = c[V(\mu) - d^2].$$

Apply (2.4) pointwise with the fixed parameter d , then average:

$$\begin{aligned} (1 + d^2)H(f | Y) &\geq c[V(\mu) - d^2] + 2cd \mathbb{E}|A - B| \\ &\geq c[V(\mu) + d^2] \\ &\geq (1 + d^2)cV(\mu). \end{aligned} \quad (2.7)$$

The last step is $cd^2[1 - V(\mu)] \geq 0$. Division by $1 + d^2$ proves the claim. $\square$

What has been gained. To prove a dimension-independent profile, it is enough to check a two-posterior inequality with the single free coefficient c . The restriction to $d \leq 1/2$ is legitimate because the splitting coordinate is chosen; it is not an assumption about every coordinate of the function.

2.5 The entropy estimate used here

Theorem 2.4 (Quantitative two-posterior mixing). *For $0 < p \leq 1/20$, every $a, b \in [0, 1]$, and every independent $d \in [0, 1/2]$,*

$$(1 + d^2)M_p(a, b) \geq \frac{h(a) + h(b)}{2} + 2dh(p)|a - b| + \frac{p}{500}(a - b)^2. \quad (2.8)$$

Section 3 proves this theorem without using Proposition 2.3. Taking $c = h(p)$ and dropping the final nonnegative term gives the profile (1.3). Keeping that term later gives the dimension-independent spectral deficit in Theorem 6.1.

2.6 All slack remains visible

Remark 2.5 (Accounting for the entire deficit). Put $\Delta_p(f) = H(f(X) | Y) - V(\mu)h(p)$, and let

$$S_{p,d}(a, b) = (1 + d^2)M_p(a, b) - \frac{h(a) + h(b)}{2} - 2dh(p)|a - b|.$$

For the sections above, direct rearrangement gives

$$\begin{aligned} (1 + d^2)\Delta_p(f) &= \mathbb{E}S_{p,d}(A, B) + \frac{\Delta_p(f_0) + \Delta_p(f_1)}{2} \\ &\quad + 2dh(p)(\mathbb{E}|A - B| - d) + d^2[1 - V(\mu)]h(p). \end{aligned} \quad (2.9)$$

The four terms are, respectively, the scalar slack, the average child deficit, the gap from the triangle inequality, and the bias surplus. Under the scalar lemma and child hypotheses, every term is nonnegative. Moreover, the scalar term is at least $(p/500)\mathbb{E}(A - B)^2$. This is the same induction, with no contribution suppressed.

The identity separates four possible sources of strictness. Neither the triangle gap nor the bias term is forced to be positive: complementary balanced sections can have $d = 0$. Section 7.2 exhibits exactly that situation.

2.7 From the profile to information

Corollary 2.6. *Once Theorem 2.4 is proved, Theorem 1.1 holds for $0 \leq p \leq 1/20$.*

Proof. Set $m = 2\mu - 1$. Then $V(\mu) = 1 - m^2$ and (1.8) gives $h(\mu) = \eta(m) \leq \ln 2 - m^2/2$. Hence

$$I(f(X); Y) \leq \ln 2 - h(p) - m^2[1/2 - h(p)] \leq \ln 2 - h(p),$$

since $h(p) \leq h(1/20) < 1/5$. At $p = 0$, use $H(f(X)) \leq \ln 2$ directly. $\square$

The use of a quadratic function of μ should not obscure the strength of the claim. When $\mu = 1/2$, the profile itself says $H(f(X) | Y) \geq h(p)$, exactly the desired conditional-entropy bound in this range. The remaining problem is to justify the scalar inequality, not to reprove an elementary variance identity.

3 The two-point entropy inequality

We now prove Theorem 2.4 independently of the induction. There are two distinct tasks: reduce arbitrary pairs to a centered comparison, and verify that comparison on its full scalar domain. The first produces a reusable entropy comparison; the second supplies constants for the chosen low-noise range. Fix p , and let

$$J = \frac{h(a) + h(b)}{2}, \quad M = M_p(a, b), \quad K = M - J, \quad t = |a - b|. \quad (3.1)$$

Here J is the entropy before mixing, M the entropy after mixing, and $K \geq 0$ the mixing gain. The scalar gap, before the extra quantitative margin is subtracted, is

$$q(d) = K + Md^2 - 2\ell d, \quad \ell = h(p)t. \quad (3.2)$$

This is a quadratic in the section-mean parameter d . Completing the square gives

$$q(d) = M \left(d - \frac{\ell}{M} \right)^2 + \frac{KM - \ell^2}{M} \quad (3.3)$$

when $M > 0$.

If $KM - \ell^2$ is nonnegative, the completed square controls every real d . Where it is negative, the restriction $0 \leq d \leq 1/2$ must be used. Both arguments benefit from the same comparison: moving a pair away from symmetry may decrease M , but it cannot decrease either K or KM . We establish that precise statement, not a separate lower bound on M .

3.1 Separation and displacement from symmetry

Swapping a, b or complementing both leaves the scalar inequality unchanged. Besides the separation t , introduce

$$\nu = |a + b - 1|, \quad 0 \leq \nu, \quad \nu + t \leq 1.$$

The pair is symmetric about $1/2$ precisely when $\nu = 0$. Define

$$Q(\nu, z) = \frac{\eta(\nu + z) + \eta(\nu - z)}{2}. \quad (3.4)$$

Then $J = Q(\nu, t)$ and $M = Q(\nu, \rho t)$. Noisy mixing preserves the midpoint of the pair and contracts its separation from t to ρt .

At the symmetric pair, write

$$J_0 = \eta(t), \quad M_0 = \eta(\rho t), \quad K_0 = M_0 - J_0. \quad (3.5)$$

These quantities correspond to $a = (1 - t)/2$ and $b = (1 + t)/2$.

Lemma 3.1 (Product centering). *For $0 < p < 1/2$ and every feasible (ν, t) ,*

$$K \geq K_0, \quad KM \geq K_0 M_0. \quad (3.6)$$

The assertion concerns the gain and a product. It does *not* say that M is minimized at symmetry, or that $q(d)$ is minimized there for each fixed d .

3.2 The curvature inequality

To compare products as the midpoint moves, we will differentiate $Q(\nu, z)^2$ in both variables. The derivative involves a difference of entropy curvatures. This suggests the coordinate $k(u) = -\eta''(u)$. Define, for $0 \leq x \leq y < 1$,

$$\mathcal{C}(x, y) = [\eta(x) + \eta(y)][k(y) - k(x)] - [\operatorname{atanh}^2 y - \operatorname{atanh}^2 x], \quad k(u) = \frac{1}{1 - u^2}. \quad (3.7)$$

For $v \geq 1$, put $g(v) = \eta(\sqrt{1 - 1/v})$. If $a = k(x)$ and $b = k(y)$, the following identity exposes the required sign:

$$\mathcal{C}(x, y) = \int_a^b [(v - a)(b - v)g''(v) + 2(g(v) + vg'(v))] dv \geq 0. \quad (3.8)$$

The proof below establishes the identity and both signs $g'' \geq 0$ and $g + vg' \geq 0$. In particular,

$$[\eta(x) + \eta(y)][k(y) - k(x)] \geq \operatorname{atanh}^2 y - \operatorname{atanh}^2 x. \quad (3.9)$$

The curvature variable is not arbitrary: $k(u) = -\eta''(u)$, exactly the factor whose difference occurs in (3.7).

Proof of the curvature identity. We prove (3.8), not merely its sign. Let $k(u) = (1 - u^2)^{-1}$ and $g(v) = \eta(u)$ with $u = \sqrt{1 - 1/v}$. The relation $v = k(u)$ gives

$$\frac{du}{dv} = \frac{(1 - u^2)^2}{2u}.$$

Using $\eta'(u) = -\operatorname{atanh} u$, differentiation yields

$$\begin{aligned} g'(v) &= -\frac{(1 - u^2)^2}{2u} \operatorname{atanh} u, \\ g''(v) &= \frac{(1 - u^2)^3}{4u^3} [(1 + 3u^2) \operatorname{atanh} u - u] \geq 0. \end{aligned} \tag{3.10}$$

The last sign uses $\operatorname{atanh} u \geq u$ for $u \geq 0$. The limits at $v = 1$ are $g'(1) = -1/2$ and $g''(1) = 5/6$.

For the other sign, write

$$g(v) + vg'(v) = \frac{w(u)}{2u}, \quad w(u) = 2u\eta(u) - (1 - u^2) \operatorname{atanh} u.$$

The continuous endpoint values of w are $w(0) = w(1) = 0$. At $u = 1$, both $\eta(u)$ and $(1 - u^2) \operatorname{atanh} u$ tend to zero. In the interior,

$$w'(u) = 2\eta(u) - 1, \quad w''(u) = -2 \operatorname{atanh} u \leq 0.$$

Thus w is concave with zero endpoints, and $w \geq 0$. At $u = 0$, $w(u)/(2u) \rightarrow \ln 2 - 1/2$. Therefore $g + vg' \geq 0$ on every finite interval $v \geq 1$.

Now fix $1 \leq a \leq b < \infty$. Integration by parts twice gives

$$\int_a^b (v - a)(b - v)g''(v) dv = (b - a)[g(a) + g(b)] - 2 \int_a^b g(v) dv. \tag{3.11}$$

For a direct verification, differentiate $(v - a)(b - v)g'(v) - (a + b - 2v)g(v)$; the result is $(v - a)(b - v)g''(v) + 2g(v)$. Its boundary values give (3.11). Also,

$$\frac{d}{dv} \operatorname{atanh}^2 \sqrt{1 - 1/v} = -2vg'(v).$$

Adding $2 \int_a^b (g + vg')$ to both sides of (3.11), and taking $a = k(x)$, $b = k(y)$, proves (3.8). Its integrand is nonnegative throughout the interval. This proof uses only finite curvature endpoints; deterministic posterior cases are obtained by continuity in the product-centering proof below.

3.3 From curvature to product centering

Proof of Lemma 3.1. First assume $\nu, t > 0$ and $\nu + t < 1$. For $0 \leq z \leq t$, define

$$L_z = -\partial_\nu Q(\nu, z) = \frac{\operatorname{atanh}(\nu + z) + \operatorname{atanh}(\nu - z)}{2} \geq 0.$$

The sign follows because Q is even and concave in ν . Differentiating with ν fixed gives

$$\partial_z L_z = \frac{1}{2}[k(\nu + z) - k(\nu - z)] \geq 0, \tag{3.12}$$

$$\partial_\nu \partial_z (Q(\nu, z)^2) = -\frac{1}{2}\mathcal{C}(|\nu - z|, \nu + z) \leq 0. \tag{3.13}$$

Therefore

$$\partial_z(Q(\nu, z)L_z) = \frac{1}{4}\mathcal{C}(|\nu - z|, \nu + z).$$

Integrating this identity from ρt to t , and recalling $J = Q(\nu, t)$ and $M = Q(\nu, \rho t)$, yields

$$JL_t - ML_{\rho t} = \frac{1}{4} \int_{\rho t}^t \mathcal{C}(|\nu - z|, \nu + z) dz. \quad (3.14)$$

Now $K = M - J$ and differentiation gives the exact identities

$$\partial_\nu K = L_t - L_{\rho t} \geq 0, \quad (3.15)$$

$$J \partial_\nu(KM) = \frac{M}{4} \int_{\rho t}^t \mathcal{C}(|\nu - z|, \nu + z) dz + K^2 L_{\rho t} \geq 0. \quad (3.16)$$

To verify the second identity directly, expand its left side as $J[ML_t - (2M - J)L_{\rho t}]$, then insert (3.14). Every term on the right of (3.16) has the asserted sign, and $J > 0$ in the interior. Hence K and KM are nondecreasing in ν . At $\nu = 0$ they equal K_0 and $K_0 M_0$.

Here are the boundary cases explicitly. If $\nu = 0$ the comparisons are equalities; if $t = 0$ both gains vanish. For $0 < t < 1$ and $\nu + t = 1$, approach the point with smaller ν and use continuity of binary entropy. If $t = 1$, feasibility forces $\nu = 0$. No derivative at a deterministic posterior and no integral to infinite curvature is used. $\square$

The two comparisons have now been proved on the whole feasible domain. We next specify when they give the desired lower bound for the quadratic in d . The proof of product centering is valid for every $0 < p < 1/2$, not just the low-noise range used in the main theorem.

3.4 Two transfer rules

Lemma 3.2 (Quantitative transfer). *For $0 < p < 1/2$ and $0 < t < 1$, let $q(d)$ be (3.2), and put*

$$q_0(d) = K_0 + M_0 d^2 - 2\ell d. \quad (3.17)$$

There are two sufficient comparisons.

If $K_0 M_0 - \ell^2 \geq \gamma \geq 0$, then, at every feasible midpoint and for every real d ,

$$q(d) \geq \frac{\gamma}{\ln 2}. \quad (3.18)$$

If $K_0 \geq D^2 M_0$, then, at every feasible midpoint and for every $0 \leq d \leq D$,

$$q(d) \geq q_0(d). \quad (3.19)$$

Proof. The first statement is product centering, the completed square (3.3), and $0 < M \leq \ln 2$.

For the second, set $w = K/K_0 \geq 1$. Product centering gives $M \geq M_0/w$. Hence

$$\begin{aligned} K + Md^2 - (K_0 + M_0 d^2) &\geq wK_0 + \frac{M_0 d^2}{w} - (K_0 + M_0 d^2) \\ &= (w - 1) \left(K_0 - \frac{M_0 d^2}{w} \right) \geq 0. \end{aligned}$$

Subtract $2\ell d$. Both factors in the last line are nonnegative because $w \geq 1$ and $d \leq D$. $\square$

The second comparison is not a global claim about fixed- d centering: it requires the stated dominance of the gain. We will establish that dominance only in the endpoint regime where it is needed.

Keep the actual denominator when it helps. For fixed separation, $Q(\nu, \rho t)$ is even and concave in ν , so $0 < M \leq M_0$ when $0 < t < 1$. If $K_0 M_0 - \ell^2 \geq 0$, product centering and completion of the square therefore give the more informative transfer

$$S_{p,d}(a, b) \geq \frac{KM - \ell^2}{M} \geq \frac{K_0 M_0 - \ell^2}{M_0} = K_0 - \frac{\ell^2}{M_0}. \quad (3.20)$$

The nonnegative numerator is essential to the second inequality. Replacing M_0 by $\ln 2$ can lose a factor of the noise scale near the endpoint; we do not do so there.

3.5 Three centered estimates and their uses

Write $\varepsilon = (1 - t)/2$, so the symmetric posteriors are $(\varepsilon, 1 - \varepsilon)$. Then $J_0 = h(\varepsilon)$ and $M_0 = h(p + \rho\varepsilon)$. There are two relevant scales: the distance ε from a deterministic pair and the noise p .

The following estimates hold throughout $0 < p \leq 1/20$. Their complete proofs are in Appendix A.

Posterior regime	Available centered estimate	Transfer used
$0 < t \leq 22/25$	$K_0 M_0 - \ell^2 \geq \frac{1999}{1250000} p t^2$	product and $M \leq \ln 2$ gain dominance
$t \geq 22/25, 0 < \varepsilon \leq p$	$q_0(d) \geq \frac{23}{3600} p t, \quad K_0 \geq M_0/3$	$0 \leq d \leq 1/2$
$t \geq 22/25, p \leq \varepsilon$	$K_0 - \ell^2/M_0 \geq \frac{453}{40000} p t$	retained denominator

The middle row is not a nonnegative-discriminant claim: its restriction on d is essential to the argument. The last row also proves that $K_0 M_0 - \ell^2 \geq 0$, which is required before changing its denominator.

Proof of Theorem 2.4. Let $0 < t < 1$. In the first regime, (3.18) gives

$$q(d) \geq \frac{1999}{1250000 \ln 2} p t^2 \geq \frac{p}{500} t^2,$$

because $1999/1250000 > 7/5000 > (\ln 2)/500$. In the second regime, $K_0 \geq M_0/3 \geq M_0/4$ permits (3.19) with $D = 1/2$. Therefore

$$q(d) \geq q_0(d) \geq \frac{23}{3600} p t \geq \frac{p}{500} t^2.$$

In the third regime use (3.20) and $453/40000 > 1/500$, again with $t \leq 1$.

The regimes cover every $0 < t < 1$, including the shared boundaries. At $t = 0$, the pair agrees and the gap is $d^2 h(a) \geq 0$. At $t = 1$, the pair is $(0, 1)$ up to swapping, so the gap is $h(p)(1 - d)^2 \geq h(p)/4 \geq p/4 \geq p/500$. Here $h(p) \geq p \ln(1/p) \geq p$ for $p \leq 1/20$. $\square$

This proves the actual posterior inequality, at every midpoint and every allowed d . Proposition 2.3 and Corollary 2.6 now apply. The positive surplus is kept for the quantitative argument of Section 6.

4 Complementary noise: direct bounds and Fourier structure

We now consider $1/20 \leq p \leq 1/2$, or equivalently $0 \leq \rho \leq 9/10$. This branch does not use the induction or extend Theorem 2.4 beyond its range. It uses the output mean, individual coordinate

correlations, and the remaining Fourier mass. The first two quantities lead to direct information estimates; only the third class needs the energy argument of Section 5.

Write

$$F = 2f - 1, \quad m = \mathbb{E}F = 2\mu - 1, \quad b_i = \mathbb{E}(FX_i). \quad (4.1)$$

The number m measures output bias, and b_i measures correlation with coordinate i . We divide functions into the following exhaustive classes:

$$\begin{aligned} |m| \geq 13/20 : & \quad \text{the output itself has sufficiently little entropy;} \\ \max_i |b_i| \geq 13/20 : & \quad \text{a coordinate comparison gives the bound;} \\ |m| \leq 13/20, \max_i |b_i| \leq 13/20 : & \quad \text{a Fourier bound handles the remainder.} \end{aligned}$$

The overlap at the boundaries is harmless. The constants 13/20 and 9/10 are used to make all three estimates fit; no optimality is asserted for this partition.

Why 13/20? The balanced lift below has coordinate coefficients at most 13/40. The interval $[7/40, 13/40]$ is symmetric about $1/4$, so the elementary section bound is at most $309/1600$, below the unchanged cap $31/160$. Meanwhile the direct coordinate estimate still has a positive endpoint margin at $b = 13/20$; Appendix B calculates it as more than $3/10000$. Thus the same cutoff is compatible with the direct estimate and the elementary Fourier estimate. It leaves no gap between the classes.

4.1 Contraction under further noise

The elementary inequality $\kappa(\theta u) \leq \theta^2 \kappa(u)$ tensorizes relative to the uniform reference measure. We state the exact form, since it will be used both conditionally and after adding further noise.

Lemma 4.1 (Uniform-reference contraction). *Let U be uniform on $\{-1, 1\}^k$, let Z be any finite random variable jointly distributed with U , and obtain U_θ by independent sign noise of correlation $\theta \in [0, 1]$, independent of (Z, U) . Then*

$$I(Z; U_\theta) \leq \theta^2 I(Z; U). \quad (4.2)$$

Proof. First allow an arbitrary law for U , and write $W = U_\theta$. Conditional on $W_{<i}$, the mean of the noisy sign W_i is θ times the mean of U_i . The second inequality in (1.8), followed by the chain rule, gives

$$\begin{aligned} k \ln 2 - H(W) &= \sum_i [\ln 2 - H(W_i | W_{<i})] \\ &\leq \theta^2 \sum_i [\ln 2 - H(U_i | W_{<i})] \\ &\leq \theta^2 [k \ln 2 - H(U)]. \end{aligned}$$

For the last inequality, $W_{<i}$ is a degraded observation of $U_{<i}$, so $H(U_i | W_{<i}) \geq H(U_i | U_{<i})$.

Apply this entropy-deficit inequality to the conditional law given each value of Z , and average. If the unconditional U is uniform, so is W . Thus the averaged deficits are exactly $I(Z; U)$ and $I(Z; W)$, proving (4.2). $\square$

Uniformity is required for this final identification with mutual information. The lemma is not a claim of the same contraction coefficient for arbitrary unconditional input distributions on the product channel.

4.2 The two direct classes

Large output bias.

Lemma 4.2. *If $|m| \geq 13/20$, then $I(f(X); Y) \leq \kappa(\rho)$ for all $0 \leq \rho \leq 1$.*

Proof. By (1.8) and $\ln 2 < 7/10$, the output entropy satisfies

$$\eta(m) \leq \ln 2 - \frac{m^2}{2} < \frac{7}{10} - \frac{169}{800} = \frac{391}{800} < \frac{1}{2}.$$

Lemma 4.1, with $Z = f(X)$ and $U = X$, therefore gives, for $\rho > 0$,

$$I(f(X); Y) \leq \rho^2 H(f(X)) < \frac{\rho^2}{2} \leq \kappa(\rho).$$

At $\rho = 0$, the observation is independent of the output. $\square$

Large coordinate correlation. A large positive coefficient $b = \mathbb{E}(FX_i)$ means that F agrees with X_i with probability $(1 + b)/2$. We exploit that coordinate directly, rather than treating its mass as part of an undifferentiated Fourier bound.

Lemma 4.3. *If some $|b_i| \geq 13/20$ and $0 \leq \rho \leq 9/10$, then $I(f(X); Y) \leq \kappa(\rho)$.*

Proof. Reverse the orientation of the coordinate if needed, so its coefficient is $b \geq 13/20$. Its section means are $(1 + m - b)/2$ and $(1 + m + b)/2$; in particular, $|m| + b \leq 1$.

Let Y_i be the noisy value of this coordinate and let X' denote all the other original coordinates. Given Y_i , the vector X' is still uniform, because the selected coordinate and its noise are independent of the remaining coordinates. Conditional on $Y_i = y_i$, the remaining noise is independent of $(f(X), X')$. Hence $f(X) \rightarrow X' \rightarrow Y'$ is a Markov chain in that conditional experiment, even though $f(X)$ may be randomized given X' . Apply Lemma 4.1 for each y_i and average. Rearranging the resulting conditional mutual-information inequality yields

$$H(f | Y) \geq (1 - \rho^2)H(f | Y_i) + \rho^2 H(f | X', Y_i). \quad (4.3)$$

The first entropy on the right is $Q(m, \rho b)$, with the even extension of (3.4) in its first variable. For the second, put $\sigma = \mathbb{P}(f_0(X') \neq f_1(X'))$. If the sections agree, the value of f is determined by X' . If they disagree, its residual entropy given Y_i is $h(p) = \eta(\rho)$. Therefore

$$H(f | X', Y_i) = \sigma \eta(\rho), \quad \sigma \geq |\mathbb{E}f_1 - \mathbb{E}f_0| = b.$$

Substitution in (4.3) gives

$$H(f | Y) \geq (1 - \rho^2)Q(m, \rho b) + \rho^2 b \eta(\rho). \quad (4.4)$$

Consequently the information is at most

$$U_\rho(m, b) = \eta(m) - (1 - \rho^2)Q(m, \rho b) - \rho^2 b \eta(\rho). \quad (4.5)$$

Two scalar facts finish the argument. On $|m| + b \leq 1$, the function $U_\rho(m, b)$ is even and concave in m , hence $U_\rho(m, b) \leq U_\rho(0, b)$. Also,

$$(1 - \rho^2)\eta(\rho b) - (1 - b\rho^2)\eta(\rho) \geq 0 \quad \left(\frac{13}{20} \leq b \leq 1, 0 \leq \rho \leq \frac{9}{10} \right). \quad (4.6)$$

Appendix B proves both facts. The last inequality is exactly $U_\rho(0, b) \leq \kappa(\rho)$. $\square$

The first reduction centers the *upper bound on information*, not the function itself. No assumption of balanced output was made in (4.4).

4.3 Fourier mass when no coordinate dominates

It remains to treat $|m|, \max_i |b_i| \leq 13/20$. For $S \subseteq \{1, \dots, n\}$, let

$$\chi_S(x) = \prod_{i \in S} x_i, \quad \hat{F}(S) = \mathbb{E}[F(X)\chi_S(X)], \quad W_k(F) = \sum_{|S|=k} \hat{F}(S)^2.$$

The characters are an orthonormal basis. Parseval gives $\sum_k W_k(F) = 1$, while $W_0(F) = m^2$ and $W_1(F) = \sum_i b_i^2$. For an indicator g , all nonconstant Fourier coefficients of $2g - 1$ are twice those of g . Their squared weights therefore differ by a factor of four.

Two elementary estimates control this class. Antipodal pairing bounds the first-level mass of a section in terms of its mean. A polynomial estimate treats linear combinations whose coefficients are all small. The mean-dependent inequality $W_1(g) \leq a/2$ is standard in substance [5, 6]; its short proof is included to fix normalization.

Lemma 4.4 (Antipodal pairing). *For any indicator g on a finite sign cube, with $a = \mathbb{E}g$, one has*

$$\frac{a}{2} - W_1(g) = \frac{1}{2} \mathbb{E}[g(X)g(-X)] + \sum_{\substack{k \geq 3 \\ k \text{ odd}}} W_k(g) \geq 0. \quad (4.7)$$

Consequently $W_1(g) \leq \min(a, 1 - a)/2$.

Proof. Set $g_{\text{odd}}(x) = [g(x) - g(-x)]/2$. Since $\chi_S(-x) = (-1)^{|S|}\chi_S(x)$, orthogonality gives

$$\mathbb{E}g_{\text{odd}}^2 = \sum_{k \text{ odd}} W_k(g).$$

On the other hand, $g^2 = g$ and the uniform law is invariant under $x \mapsto -x$, so

$$\mathbb{E}g_{\text{odd}}^2 = \frac{1}{4} \mathbb{E}[g(X) - g(-X)]^2 = \frac{a}{2} - \frac{1}{2} \mathbb{E}[g(X)g(-X)].$$

Separating degree one proves (4.7). Applying the bound to $1 - g$ gives the complementary mean bound, because nonconstant squared coefficients are unchanged. $\square$

For balanced functions, this elementary bound becomes useful after splitting at a coordinate with a moderately large coefficient.

For the remaining small-coefficient case, we need the following bound on a sum of independent signs. Its constant is sufficient for the Fourier cap below.

Lemma 4.5 (A capped sign-sum estimate). *Let $R = \sum_j a_j X_j$ for independent uniform signs, with $\sum_j a_j^2 = 1$ and $\max_j |a_j| \leq 2/5$. Then*

$$\mathbb{E}|R| \leq \frac{7}{8}. \quad (4.8)$$

Appendix C gives a fixed even polynomial $P \geq |\cdot|$ by a factored identity, computes its moments, and reduces $\mathbb{E}P(R) \leq 7/8$ to two rational endpoint comparisons. The polynomial degree is not a restriction on the Fourier degree of g or on the number of inputs.

Lemma 4.6 (A balanced first-level estimate). *If g is balanced and $\max_i |\mathbb{E}(gX_i)| \leq 13/40$, then $W_1(g) \leq 31/160$.*

Proof. Put $c_i = \mathbb{E}(gX_i)$, $\beta = \max_i |c_i|$, $w = \sum_i c_i^2$, and $L_1 = \sum_i c_i X_i$. Since $0 \leq g \leq 1$ and L_1 is symmetric,

$$w = \mathbb{E}(gL_1) \leq \mathbb{E}(L_1)_+ = \frac{1}{2}\mathbb{E}|L_1|. \quad (4.9)$$

First suppose $\beta \leq 7/40$. If $w \geq 31/160$, the coefficients of $L_1/\sqrt{w}$ have squared maximum at most

$$\frac{(7/40)^2}{31/160} = \frac{49}{310} < \frac{4}{25}.$$

Lemma 4.5 gives $\mathbb{E}|L_1| \leq (7/8)\sqrt{w}$. Equation (4.9) then implies

$$w \leq \frac{7}{16}\sqrt{w}, \quad w \leq \frac{49}{256} < \frac{31}{160},$$

a contradiction. Thus $w < 31/160$ in this case.

For $7/40 \leq \beta \leq 13/40$, split at a positively oriented coordinate attaining β . The section means are $1/2 - \beta$ and $1/2 + \beta$. Apply Lemma 4.4 to the sections, complementing the second. Complementation preserves coefficient norms, so each original section vector has squared norm at most $(1/2 - \beta)/2$. Their average is the parent vector; the triangle inequality gives

$$w \leq \beta^2 + \frac{1/2 - \beta}{2} = \left(\beta - \frac{1}{4}\right)^2 + \frac{3}{16} \leq \left(\frac{3}{40}\right)^2 + \frac{3}{16} = \frac{309}{1600} < \frac{31}{160}. \quad (4.10)$$

Here $|\beta - 1/4| \leq 3/40$ is exactly $\beta \in [7/40, 13/40]$. Together with the small- β case it covers every permitted coefficient. □

Including the output mean. The estimate just proved assumes balance. We need an estimate for $m^2 + W_1(F)$ with no balance assumption. The following lift puts m into one extra linear coefficient and therefore preserves exactly the quantity needed, not mutual information.

Proposition 4.7 (Bias-inclusive Fourier estimate). *If $|m|, \max_i |b_i| \leq 13/20$, then*

$$m^2 + W_1(F) \leq \omega, \quad \omega := \frac{31}{40}. \quad (4.11)$$

Proof. Add an independent uniform sign z and set

$$F^\sharp(x, z) = zF(zx).$$

With the change of variables $u = zx$, the pair (u, z) is independent and uniform. It follows that

$$\mathbb{E}F^\sharp = 0, \quad \mathbb{E}(F^\sharp z) = \mathbb{E}F(u) = m, \quad \mathbb{E}(F^\sharp x_i) = \mathbb{E}(F(u)u_i) = b_i.$$

Thus the indicator $(1 + F^\sharp)/2$ is balanced and its coordinate coefficients have magnitude at most $13/40$. Lemma 4.6 gives

$$\frac{m^2 + \sum_i b_i^2}{4} \leq \frac{31}{160},$$

which is (4.11). □

The lift is used only for the coefficient estimate. We do not identify the mutual information of $F^\sharp$ with that of F .

5 From Fourier structure to the information bound

For the remaining class, we now know that the constant and linear Fourier levels carry at most $\omega = 31/40$ of the total squared mass. The next task is to turn that geometric fact into a lower bound on conditional entropy.

5.1 Degree-two cancellation

The noise operator is

$$T_\rho F(y) = \mathbb{E}[F(X) \mid Y = y].$$

Since $\mathbb{E}(X_i \mid Y_i) = \rho Y_i$, coordinate independence gives $T_\rho \chi_S = \rho^{|S|} \chi_S$. Set $G = T_\rho F$ and $R = |G|$. The posterior probability of $f(X) = 1$ is $(1 + G(Y))/2$, hence

$$\mathbb{H}(f(X) \mid Y) = \mathbb{E}\eta(R). \quad (5.1)$$

Here and below, expectations of products of cube functions such as FG use the *same uniform argument*. In particular $\mathbb{E}(FG)$ means $\mathbb{E}[F(U)G(U)]$ for uniform U ; it does not mean $\mathbb{E}[F(X)G(Y)]$ at the two correlated channel endpoints. The entropy identity is unchanged because U and Y are both uniform.

A bound on $\mathbb{E}G^2$ alone would retain contributions from every Fourier level. Instead, consider

$$\mathbb{E}G^2 - \rho^2 \mathbb{E}(FG) = \sum_{k \geq 0} (\rho^{2k} - \rho^{k+2}) W_k(F). \quad (5.2)$$

The factor ρ^2 is chosen so that the degree-two coefficient vanishes. The degree-zero and degree-one coefficients are nonnegative, and every coefficient of degree at least three is nonpositive. Thus the higher levels can be discarded in the direction we need:

$$\mathbb{E}G^2 - \rho^2 \mathbb{E}(FG) \leq (1 - \rho^2)m^2 + \rho^2(1 - \rho)W_1(F). \quad (5.3)$$

The Boolean constraint $|F| = 1$ gives the pointwise inequality $FG \leq |G| = R$.

5.2 The posterior statistic and the entropy comparison

Rearranging (5.3) produces $1 - \rho^2 + \rho^2 R - R^2$. This is why the comparison function is

$$\psi_\rho(t) = (1 - t)(1 + t - \rho^2), \quad 0 \leq t \leq 1. \quad (5.4)$$

It is nonnegative on this interval and vanishes when the posterior is certain ($t = 1$), as entropy does. Put

$$\Lambda_\rho = 1 + \rho - \omega\rho^2, \quad \Gamma_\rho = 1 + \rho - \rho^2. \quad (5.5)$$

Both are positive on $[0, 1]$.

Lemma 5.1 (Energy identity and its nonnegative remainder). *For $0 < \rho < 1$, define*

$$\mathcal{T}_\rho(F) = \sum_{k=3}^n (\rho^{k+2} - \rho^{2k}) W_k(F), \quad s = \omega - m^2 - W_1(F).$$

Then the exact identity is

$$\begin{aligned} & \mathbb{E}\psi_\rho(R) - (1 - \rho)(\Lambda_\rho - \Gamma_\rho m^2) \\ &= \rho^2 \mathbb{E}(R - FG) + \mathcal{T}_\rho(F) + \rho^2(1 - \rho)s. \end{aligned} \quad (5.6)$$

Under (4.11), all three terms on the right are nonnegative. In particular,

$$\mathbb{E}\psi_\rho(R) \geq (1 - \rho)(\Lambda_\rho - \Gamma_\rho m^2). \quad (5.7)$$

Proof. Use the exact spectral equality (5.2), rather than its upper bound. The degree-zero and degree-one contributions are $(1 - \rho^2)m^2$ and $\rho^2(1 - \rho)W_1(F)$; degree two is zero. Thus

$$\mathbb{E}G^2 = \rho^2\mathbb{E}(FG) + (1 - \rho^2)m^2 + \rho^2(1 - \rho)W_1(F) - \mathcal{T}_\rho(F).$$

Insert this in $\mathbb{E}\psi_\rho(R) = 1 - \rho^2 + \rho^2\mathbb{E}R - \mathbb{E}G^2$, and write $W_1(F) = \omega - m^2 - s$. Rearranging gives (5.6).

For the signs, $R - FG \geq 0$ pointwise because $F \in \{-1, 1\}$. For $k \geq 3$, $k + 2 \leq 2k$ and $0 < \rho < 1$, hence $\rho^{k+2} - \rho^{2k} \geq 0$. Finally $s \geq 0$ is exactly (4.11). There is no unestimated Fourier tail. The endpoints extend by continuity. $\square$

The task is now scalar again: find a multiple of ψ_ρ below η . For a balanced output, (5.7) supplies $(1 - \rho)\Lambda_\rho$. To obtain the desired entropy $\eta(\rho)$, the natural coefficient is therefore

$$\frac{\eta(\rho)}{(1 - \rho)\Lambda_\rho}.$$

The coefficient is determined by the target, not chosen independently of it.

5.3 A reusable Fourier-to-entropy certificate

The preceding mechanism can be stated without the particular cap $31/40$. This isolates two inputs that can be improved independently: a Fourier cap and a scalar lower envelope for entropy.

Proposition 5.2 (Information from a spectral cap and an entropy envelope). *Fix $0 < \rho < 1$ and a sign-valued Boolean F , and write $f = (1 + F)/2$, $m = \mathbb{E}F$, $R = |T_\rho F|$. Suppose*

$$m^2 + W_1(F) \leq \Omega \leq 1, \quad \eta(t) \geq \lambda\psi_\rho(t) + \zeta(1 - t) \quad (0 \leq t \leq 1),$$

where $\lambda, \zeta \geq 0$. Put $\Lambda_\Omega = 1 + \rho - \Omega\rho^2$ and $\Gamma = 1 + \rho - \rho^2$. Then

$$\begin{aligned} \kappa(\rho) - \mathsf{I}(f; Y) &\geq \lambda(1 - \rho)\Lambda_\Omega - \eta(\rho) \\ &\quad + m^2 \left[\frac{1}{2} - \lambda(1 - \rho)\Gamma \right] + \zeta\mathbb{E}(1 - R). \end{aligned} \quad (5.8)$$

In particular, the CK bound follows for this F whenever

$$\lambda(1 - \rho)\Lambda_\Omega \geq \eta(\rho), \quad \lambda(1 - \rho)\Gamma \leq \frac{1}{2}. \quad (5.9)$$

Proof. The energy identity is algebraic in the cap. With $s_\Omega = \Omega - m^2 - W_1(F) \geq 0$, its exact form is

$$\mathbb{E}\psi_\rho(R) - (1 - \rho)(\Lambda_\Omega - \Gamma m^2) = \rho^2\mathbb{E}(R - FG) + \mathcal{T}_\rho(F) + \rho^2(1 - \rho)s_\Omega \geq 0,$$

where $G = T_\rho F$ and the products use the same uniform argument. Multiply by $\lambda \geq 0$, average the scalar envelope, and use $\mathsf{H}(f) = \eta(m) \leq \ln 2 - m^2/2$. Subtracting the resulting information upper bound from $\kappa(\rho)$ gives (5.8). $\square$

The first condition in (5.9) pays for the balanced target. The second pays for the loss caused by a nonzero mean. A nonzero ζ leaves an explicit stability remainder. This proposition requires no balance assumption; all dependence on m has been displayed.

5.4 The envelope needed for this proof

We use $\Omega = 31/40$. The following two bounds supply the scalar envelopes and the strict starting point for adding further noise.

Lemma 5.3 (Entropy comparisons). *For all $t \in [0, 1]$,*

$$\eta(t) \geq \frac{\eta(\rho)}{(1-\rho)\Lambda_\rho} \psi_\rho(t) \quad \left(\frac{3}{5} \leq \rho \leq \frac{9}{10} \right), \quad (5.10)$$

$$\eta(t) \geq \frac{39}{40} \psi_{3/5}(t). \quad (5.11)$$

Appendix D proves these inequalities. The proof of (5.10) constructs a piecewise-affine upper bound for the required coefficient, then uses entropy concavity and three positive polynomial expansions. The second inequality is a deliberately stronger estimate at one correlation, needed to continue toward still higher noise.

5.5 Accounting for output bias

Proposition 5.4. *Under $|m|, \max_i |b_i| \leq 13/20$, the target holds for $3/5 \leq \rho \leq 9/10$.*

Proof. Combine (5.1), (5.7), and (5.10):

$$H(f(X) | Y) \geq \eta(\rho) \left(1 - \frac{\Gamma_\rho}{\Lambda_\rho} m^2 \right).$$

Meanwhile $H(f(X)) = \eta(m) \leq \ln 2 - m^2/2$. Subtracting gives

$$I(f(X); Y) \leq \kappa(\rho) + m^2 \left[\frac{\eta(\rho)\Gamma_\rho}{\Lambda_\rho} - \frac{1}{2} \right]. \quad (5.12)$$

The bracket is negative. The factor $\eta(\rho)$ decreases, and so does

$$\frac{\Gamma_\rho}{\Lambda_\rho} = 1 - (1-\omega) \frac{\rho^2}{\Lambda_\rho}, \quad \left(\frac{\rho^2}{\Lambda_\rho} \right)' = \frac{\rho(2+\rho)}{\Lambda_\rho^2} > 0.$$

At $\rho = 3/5$, the two positive factors satisfy

$$\eta(3/5) \leq \ln 2 - \frac{9}{50} < \frac{13}{25}, \quad \frac{\Gamma_{3/5}}{\Lambda_{3/5}} = \frac{1240}{1321}, \quad \frac{13}{25} \frac{1240}{1321} < \frac{1}{2}.$$

Thus the bias term in (5.12) is nonpositive throughout the interval. $\square$

This is the explicit reason the balanced comparison extends to arbitrary means. The conditional-entropy estimate loses a term proportional to m^2 , but the smaller output entropy pays for that loss.

5.6 Why a strict anchor is needed for higher noise

For $0 < \rho < \rho_0 = 3/5$, adding noise to only the bound $l \leq \kappa(\rho_0)$ gives an upper bound strictly *larger* than the target. Indeed, the positive series (1.7) gives

$$\frac{\kappa(u)}{u^2} = \sum_{j \geq 1} \frac{u^{2j-2}}{(2j)(2j-1)},$$

which is strictly increasing for $0 < u < 1$. Consequently $(\rho/\rho_0)^2 \kappa(\rho_0) > \kappa(\rho)$. We therefore establish an anchor strictly below $\rho_0^2/2$ before degrading. At $\rho = 0$, both quantities vanish.

Proposition 5.5. *Under $|m|, \max_i |b_i| \leq 13/20$, the target holds for $0 \leq \rho \leq 3/5$.*

Proof. Let $\rho_0 = 3/5$ and let Y_0 be the observation at that correlation. Apply (5.11) to (5.7). Since

$$\Lambda_{\rho_0} = \frac{1321}{1000}, \quad \Gamma_{\rho_0} = \frac{31}{25},$$

we obtain

$$\begin{aligned} I(f(X); Y_0) &\leq \ln 2 - \frac{39}{40} \frac{2}{5} \frac{1321}{1000} + m^2 \left[\frac{39}{40} \frac{2}{5} \frac{31}{25} - \frac{1}{2} \right] \\ &\leq U_0 := \ln 2 - \frac{39}{40} \frac{2}{5} \frac{1321}{1000} < \frac{9}{50} = \frac{\rho_0^2}{2}. \end{aligned}$$

The bracket is negative; the last finite comparison is in Appendix E.

For $0 < \rho \leq \rho_0$, pass Y_0 through independent sign noise of correlation ρ/ρ_0 . The resulting vector has the same joint distribution with $f(X)$ as the observation Y at correlation ρ . Since Y_0 is uniform, Lemma 4.1 gives

$$I(f(X); Y) \leq \left(\frac{\rho}{\rho_0} \right)^2 U_0 < \frac{\rho^2}{2} \leq \kappa(\rho).$$

Only degradation toward more noise is used. At $\rho = 0$, independence gives zero mutual information. $\square$

5.7 Proof of the full-range theorem

Proof of Theorem 1.1. For $0 < p \leq 1/20$, Theorem 2.4, Proposition 2.3, and Corollary 2.6 prove the target.

For $1/20 \leq p < 1/2$, set $\rho = 1 - 2p \in (0, 9/10]$. If $|m| \geq 13/20$, use Lemma 4.2. If a coordinate coefficient has magnitude at least $13/20$, use Lemma 4.3. Otherwise Proposition 4.7 applies, and Propositions 5.4 and 5.5 give the result. These cases cover every Boolean function and every output mean.

At $p = 0$, the claim is $H(f(X)) \leq \ln 2$. At $p = 1/2$, Y is independent of X , so the mutual information is zero. The two nontrivial ranges meet at $p = 1/20$. $\square$

This completes the proof of Theorem 1.1. The two branches use different estimates but the same entropy–information identity. The next section is not needed for that conclusion: it keeps quantitative slack to describe how close a nearly optimal function must be to a coordinate.

6 Spectral deficit and stability

The core CK inequality is complete. We now keep the scalar surplus instead of discarding it. The point is not a larger numerical margin alone: it can be propagated without losing a factor depending on dimension.

The induction below uses only a bound $S_{p,d}(a, b) \geq \lambda(a - b)^2$. Consequently its proof also yields the reusable implication

$$S_{p,d} \geq \lambda(a - b)^2 \text{ on the full posterior domain} \implies \Delta_p(f) \geq \lambda \rho^2 \sum_{k \geq 2} W_k(F), \quad (6.1)$$

for $0 < p < 1/2$ and $\lambda \geq 0$, provided the scalar bound is valid for independently all $0 \leq d \leq 1/2$. The displayed theorem specializes this implication to the coefficient established above.

6.1 A second induction: only the lost spectral mass matters

Let $F = 2f - 1$, $m = \mathbb{E}F$, and put

$$\mathcal{E}(F) = \sum_{k \geq 2} W_k(F) = 1 - m^2 - W_1(F).$$

This is the squared Fourier mass above degree one. A bound based on the smallest eigenvalue of the $(n - 1)$ -coordinate noise operator would contain $\rho^{2(n-1)}$. Instead, the induction tracks only the degree-two mass lost by the chosen restriction. The posterior difference restores exactly that loss.

Theorem 6.1 (Dimension-free spectral deficit). *For every finite dimension and $0 < p \leq 1/20$,*

$$\Delta_p(f) := \mathbb{H}(f(X) \mid Y) - 4\mu(1 - \mu)h(p) \geq \frac{p\rho^2}{500} \mathcal{E}(F), \quad \rho = 1 - 2p. \quad (6.2)$$

Proof. Set $\lambda = p/500$. Induct on dimension, with the same small-gap coordinate as in Proposition 2.3. The zero-bit and one-bit cases have $\mathcal{E}(F) = 0$ and satisfy the bound directly. Let i be the selected coordinate, $d = |\hat{F}(\{i\})| \leq 1/2$, and define

$$T_i = \sum_{j \neq i} \hat{F}(\{i, j\})^2.$$

For the two sign-valued sections F_0, F_1 , their coefficients give the exact identity

$$\frac{\mathcal{E}(F_0) + \mathcal{E}(F_1)}{2} = \mathcal{E}(F) - T_i. \quad (6.3)$$

Indeed, the averaged squared mean is $m^2 + d^2$, and the averaged first-level mass is $W_1(F) - d^2 + T_i$.

For the common-observation posteriors, $B - A = T_\rho(f_1 - f_0)$. The Fourier coefficients of $f_1 - f_0$ are exactly $\hat{F}(S)$ for $S \ni i$, with i removed from the index. Orthogonality yields

$$\mathbb{E}(A - B)^2 = \sum_{S \ni i} \rho^{2(|S|-1)} \hat{F}(S)^2 \geq d^2 + \rho^2 T_i. \quad (6.4)$$

Use the exact deficit identity (2.9), the strengthened scalar bound, and the induction hypothesis for the sections. Its other terms are nonnegative, so

$$\begin{aligned} (1 + d^2)\Delta_p(f) &\geq \lambda \rho^2 [\mathcal{E}(F) - T_i] + \lambda [d^2 + \rho^2 T_i] \\ &= \lambda \rho^2 \mathcal{E}(F) + \lambda d^2 \\ &\geq (1 + d^2) \lambda \rho^2 \mathcal{E}(F). \end{aligned}$$

The last inequality is $\rho^2 \mathcal{E}(F) \leq 1$. Divide by $1 + d^2$. No lower bound on the smallest eigenvalue in dimension n is used. $\square$

The same scalar improvement also gives the one-coordinate estimate

$$\Delta_p(f) \geq \frac{p}{500(1+d^2)} \mathbb{E}(A-B)^2 \geq \frac{p}{625} \rho^{2(n-1)} \mathbb{P}(f_0 \neq f_1). \quad (6.5)$$

That coordinate estimate still depends on n ; Theorem 6.1 does not. The two statements measure different aspects of the function and are not identified with each other.

6.2 From spectral mass to distance from a coordinate

For $n \geq 1$, let

$$b = \max_i |\mathbb{E}(FX_i)|, \quad \delta(f) = \min_{i, \sigma \in \{-1, 1\}} \mathbb{P}(F(X) \neq \sigma X_i) = \frac{1-b}{2}.$$

To turn a spectral deficit into a statement about functions, we need a bound on distance to a coordinate. The following nonsharp estimate follows from Section 4. Its role is the same as a first-level stability theorem [7], but the constants needed here follow directly from the estimates already proved.

Lemma 6.2 (An explicit distance bound). *For every Boolean f on at least one input,*

$$\delta(f) \leq \frac{20}{9} [m^2 + \mathcal{E}(F)] = \frac{20}{9} [1 - W_1(F)]. \quad (6.6)$$

Proof. If $|m|, b \leq 13/20$, Proposition 4.7 gives $\mathcal{E}(F) \geq 9/40$, and $\delta(f) \leq 1/2$ proves the claim. If $|m| \geq 13/20$, use $\delta(f) \leq 1/2 < (20/9)(13/20)^2 \leq (20/9)m^2$.

It remains that $b \geq 13/20$. Apply the balancing lift $F^\sharp(x, z) = zF(zx)$, so $W_1(F^\sharp) = m^2 + W_1(F)$. For a balanced sign function with a coordinate coefficient b , restriction into sections and Lemma 4.4 give

$$W_1(F^\sharp) \leq b^2 + 1 - b.$$

To check the factors, the corresponding indicator has coefficient $\beta = b/2$ and section means $1/2 \pm \beta$. The same section-vector argument as in (4.10), before imposing its interval cutoff, gives $W_1((1 + F^\sharp)/2) \leq \beta^2 + (1/2 - \beta)/2$ for all $0 \leq \beta \leq 1/2$. Multiplication by four is the displayed sign-output bound. Hence

$$\mathcal{E}(F) \geq b(1-b) \geq \frac{13}{10} \delta(f),$$

which is stronger than (6.6). □

Combining Theorem 6.1 with the explicit output-bias adjustment gives the useful low-noise conclusion

$$\kappa(\rho) - \mathbb{I}(f(X); Y) \geq \frac{9p\rho^2}{10000} \delta(f), \quad 0 < p \leq 1/20. \quad (6.7)$$

Indeed,

$$\kappa(\rho) - \mathbb{I}(f; Y) \geq \Delta_p(f) + m^2[1/2 - h(p)] \geq \frac{p\rho^2}{500} [\mathcal{E}(F) + m^2],$$

because $h(p) < 1/5$ and $p\rho^2/500 \leq 1/10000 < 3/10$. Now use Lemma 6.2.

6.3 A stability bound throughout the nontrivial noise range

Theorem 6.3 (Full-range quantitative consequence). *For every $n \geq 1$, every Boolean f , and every $0 \leq p \leq 1/2$,*

$$\ln 2 - h(p) - I(f(X); Y) \geq \frac{p(1-2p)^2}{2000} \delta(f). \quad (6.8)$$

Proof. Equation (6.7) proves the result for $0 < p \leq 1/20$. For the complementary range we establish the stronger bound

$$\kappa(\rho) - I(f; Y) \geq \frac{\rho^2}{2000} \delta(f), \quad 0 \leq \rho \leq 9/10, \quad (6.9)$$

using the complementary estimates with cutoff $13/20$.

Large bias. Lemma 4.2 gives $H(f) < 391/800$, so uniform-reference contraction yields a gap at least $9\rho^2/800$, which exceeds the right side of (6.9).

Large coordinate. Write $b = \max_i |\mathbb{E}(FX_i)| \geq 13/20$. The argument of Lemma 4.3 bounds the gap below by

$$G_\rho(b) = (1 - \rho^2)\eta(\rho b) - (1 - b\rho^2)\eta(\rho).$$

Concavity in b and $G_\rho(1) = 0$ give

$$G_\rho(b) \geq \frac{40}{7} \delta(f) G_\rho(13/20).$$

Appendix B proves that $A(x) = G_{\sqrt{x}}(13/20)/[x(1-x)]$ extends continuously to zero and is concave on $[0, 81/100]$. Its endpoint estimates are

$$A(0) = \frac{7}{20} \left(\frac{33}{40} - \ln 2 \right) > \frac{7}{160}, \quad A(81/100) > \frac{3/10000}{(81/100)(19/100)} = \frac{1}{513}.$$

Therefore $G_\rho(13/20) \geq \rho^2(1 - \rho^2)/513 \geq \rho^2/2700$. This gives a gap at least $2\rho^2\delta(f)/945$, stronger than (6.9).

Remainder class, $3/5 \leq \rho \leq 9/10$. Every Bernstein coefficient in Appendix D is at least $s_* = 273/25000$. Retaining this minimum in its proof gives

$$\eta(t) \geq \frac{\eta(\rho)}{(1 - \rho)\Lambda_\rho} \psi_\rho(t) + \frac{s_*}{2}(1 - t). \quad (6.10)$$

Specifically, that proof gives $\eta(t) - \lambda\psi_\rho(t) \geq yP_j(z)$ with $y = (1 - t)/2$, $P_j \geq s_*$, and λ no smaller than the required coefficient. Since $\psi_\rho \geq 0$, decreasing λ preserves the estimate.

On this correlation interval, $\Lambda_\rho \geq 5/4$ by concavity and its endpoint values, whereas $\Gamma_\rho \leq 5/4$. In the remainder class $m^2 \leq 169/400$, so $\Lambda_\rho - \Gamma_\rho m^2 > 2/3$. Put $R = |T_\rho F|$. The energy bound and $\psi_\rho(R) \leq (2 - \rho^2)(1 - R)$ imply

$$\mathbb{E}(1 - R) \geq \frac{(1 - \rho)(\Lambda_\rho - \Gamma_\rho m^2)}{2 - \rho^2} \geq \frac{5}{123} > \frac{1}{25}.$$

Subtract the average of (6.10) from $H(f) \leq \ln 2 - m^2/2$. The bias term is nonnegative in the information gap by Proposition 5.4. Consequently the gap is at least

$$\frac{273}{1250000} > \frac{81}{400000} \geq \frac{\rho^2\delta(f)}{2000}.$$

Remainder class, $0 < \rho \leq 3/5$. The strict anchor in Proposition 5.5 satisfies

$$U_0 = \ln 2 - \frac{39}{40} \frac{2}{5} \frac{1321}{1000} < \frac{179}{1000}.$$

This slightly sharper finite comparison is certified by the same Appendix E enclosure. Degradation gives a gap at least

$$\rho^2 \left[\frac{1}{2} - \frac{U_0}{(3/5)^2} \right] \geq \frac{\rho^2}{360} \geq \frac{\rho^2 \delta(f)}{2000}.$$

These cases prove (6.9). It implies (6.8) because $p \leq 1$. At $p = 0$, the right side vanishes and $H(f) \leq \ln 2$; at $p = 1/2$, both sides vanish by independence. $\square$

For any fixed $0 < p < 1/2$, Theorem 6.3 says that an information gap at most ε forces

$$\delta(f) \leq \min \left\{ \frac{1}{2}, \frac{2000\varepsilon}{p(1-2p)^2} \right\}.$$

The coefficient is independent of input dimension and is not asserted optimal. Equality in the CK bound at nontrivial noise therefore forces a coordinate indicator or its complement. The factor vanishes at both trivial endpoints, where additional equality cases are unavoidable.

Scope of the stability statement. The coefficient is uniform in the number of inputs but tends to zero as p approaches either trivial endpoint. The results quantify the same inequality; they do not require an additional noise restriction, an external stability theorem, or a numerical certificate.

7 Using the conclusions and locating their limits

The preceding proofs provide both a bound to apply and intermediate statements to reuse. This section distinguishes those uses from numerical evidence or an assertion of optimal constants.

7.1 An explicit route from information to a coordinate

Throughout this section, $n \geq 1$. Let

$$\mathcal{D}_p(f) = \ln 2 - h(p) - I(f(X); Y).$$

For $0 < p < 1/2$, the proof gives the stronger piecewise coefficient

$$c_p = \begin{cases} \frac{9p(1-2p)^2}{10000}, & 0 < p < 1/20, \\ \frac{(1-2p)^2}{2000}, & 1/20 \leq p < 1/2, \end{cases} \quad \mathcal{D}_p(f) \geq c_p \delta(f). \quad (7.1)$$

These are precisely (6.7) and (6.9); at the shared endpoint we use the larger coefficient. The change of formula reflects the proof, not a discontinuity of mutual information.

Corollary 7.1 (Recovering a coordinate from a near-optimal value). *If $\mathcal{D}_p(f) \leq \epsilon$ at fixed $0 < p < 1/2$, there are i and $\sigma \in \{-1, 1\}$ such that*

$$\mathbb{P}\{2f(X) - 1 = \sigma X_i\} \geq 1 - \min\{1/2, \epsilon/c_p\}. \quad (7.2)$$

For $\epsilon \geq 0$, the coordinate can be chosen to maximize $|\mathbb{E}[(2f - 1)X_i]|$, with its sign chosen to agree with that coefficient.

Proof. For $F = 2f - 1$, $\mathbb{P}(F \neq \sigma X_i) = (1 - \sigma \mathbb{E}(FX_i))/2$. Minimizing this expression identifies the stated coordinate and gives $\delta(f) = (1 - \max_i |\mathbb{E}(FX_i)|)/2$. Now use (7.1) and $\delta(f) \leq 1/2$. $\square$

For example, at $p = 1/4$ one has $c_p = 1/8000$. An information gap of at most 10^{-6} nats therefore guarantees disagreement on at most $1/125$ of the inputs, in every finite dimension. The coefficient is deliberately conservative. The corollary becomes nontrivial only when a valid upper bound on the information gap is sufficiently small. It neither supplies a statistical estimator of that gap nor claims sharp numerical performance or an efficient learning algorithm.

There is also an elementary converse, requiring no new CK machinery.

Proposition 7.2 (Closeness implies a small information gap). *For $0 < p < 1/2$,*

$$c_p \delta(f) \leq \mathcal{D}_p(f) \leq h(\delta(f)). \quad (7.3)$$

Proof. The lower bound is (7.1). Choose a nearest coordinate indicator g and put $E = f \oplus g$, the error bit. The chain rule gives

$$\mathbb{I}(g; Y) - \mathbb{I}(f; Y) = \mathbb{I}(g; Y | f) - \mathbb{I}(f; Y | g) \leq \mathbb{H}(g | f) = \mathbb{H}(E | f) \leq h(\delta(f)).$$

Since $\mathbb{I}(g; Y) = \ln 2 - h(p)$, this is the upper bound. $\square$

The two bounds give a qualitative equivalence between near optimality and nearness to a coordinate at fixed nontrivial noise. Their rates do not match: the lower bound is linear, whereas $h(\delta)$ is of order $\delta \ln(1/\delta)$ near zero. No sharp stability rate is asserted.

7.2 Three examples that explain the hypotheses

Constants and coordinates. A constant output has $V(\mu) = 0$, zero conditional entropy, and zero information. It saturates the variance profile but not the information optimum when $0 < p < 1/2$. A coordinate has $V = 1$ and conditional entropy $h(p)$, and saturates both bounds. This distinction explains the separate output-bias term in Corollary 2.6.

Parity: zero mean gap does not mean zero posterior gap. For $F(x) = \prod_{i=1}^k x_i$, the noisy posterior mean is $T_\rho F = \rho^k F$. Thus

$$\mathbb{H}(f | Y) = \eta(\rho^k), \quad \Delta_p(f) = \mathcal{D}_p(f) = \eta(\rho^k) - \eta(\rho). \quad (7.4)$$

For $k \geq 2$, all mass is at degree k , and $\delta(f) = 1/2$. The two-bit case is especially useful: both sections are balanced, so $d = 0$, but their posterior difference has constant magnitude ρ . In the exact deficit identity (2.9), the two child deficits, the triangle term, and the bias surplus vanish. All remaining entropy comes from the scalar mixing slack.

AND: unequal section means are unavoidable. For $f(x) = \mathbf{1}\{x_1 = x_2 = 1\}$,

$$F(x) = \frac{-1 + x_1 + x_2 + x_1 x_2}{2}, \quad \mu = \frac{1}{4}, \quad m^2 = \frac{1}{4}, \quad W_1(F) = \frac{1}{2}, \quad \sum_{k \geq 2} W_k(F) = \frac{1}{4}, \quad \delta(f) = \frac{1}{4}.$$

The conditional entropy can be calculated without Fourier theory:

$$\mathbb{H}(f | Y) = \frac{1}{4} [h(p^2) + 2h(p(1-p)) + h((1-p)^2)]. \quad (7.5)$$

The four terms correspond to the four equally likely noisy observations. Fixing one coordinate gives a constant section and a coordinate section, not two balanced functions. The example checks both the meaning of the variance loss and the factor-four conversion between indicator and sign Fourier weights.

7.3 Why the uniform scalar margin has the scale it does

Proposition 7.3 (An upper limit on a uniform quadratic surplus). *Fix $0 < p < 1/2$. If a number $\gamma(p)$ satisfies $S_{p,d}(a, b) \geq \gamma(p)(a - b)^2$ for all $a, b \in [0, 1]$ and independently all $d \in [0, 1/2]$, then*

$$\gamma(p) \leq 2p(1 - p). \quad (7.6)$$

Proof. Take $d = 0$, $a = (1 - t)/2$, and $b = (1 + t)/2$. The slack is then $\eta(\rho t) - \eta(t)$. Since $\eta'(0) = 0$ and $\eta''(0) = -1$, Taylor's formula gives

$$\lim_{t \downarrow 0} \frac{\eta(\rho t) - \eta(t)}{t^2} = \frac{1 - \rho^2}{2} = 2p(1 - p).$$

Apply the hypothesized bound and take this limit. □

For $0 < p \leq 1/20$, the best possible coefficient in this particular all-posterior statement therefore lies between $p/500$ and $2p(1 - p)$. In particular it cannot be bounded below by $Cp \ln(1/p)$ for one positive constant C and all sufficiently small p .

This does not preclude a logarithm for actual posteriors attached to a fixed Boolean function. For two-bit parity, (7.4) reads

$$\Delta_p(f) = h(2p(1 - p)) - h(p) = p \ln(1/p) + (1 - 2 \ln 2)p + O(p^2 \ln(1/p)).$$

Here the posteriors approach deterministic endpoints as $p \downarrow 0$. The uniform scalar coefficient and the deficit along such a family of posteriors are different objects. This fixed two-bit expansion is not a uniform asymptotic theorem over dimension.

7.4 What can be taken from the proof

Proposition 2.3 can be used whenever a two-posterior inequality supplies a profile coefficient c . Lemma 3.1 supplies an entropy-product comparison without reference to a Boolean dimension. Proposition 5.2 separates a spectral cap from the entropy envelope needed to convert it into information. The induction behind (6.1) explains how a local surplus pays for precisely the spectral mass lost under restriction.

These statements suggest concrete questions rather than a blanket claim of generality. How large can the uniform scalar surplus be within the upper limit (7.6)? Can a better cap or a better envelope improve the constants in (5.8)? What additional statistics of realizable posteriors recover information discarded by the all-pair scalar bound? None of the present results covers biased product inputs or asymmetric channels: equal section weights, uniform output of the channel, and the sign-cube Fourier basis enter explicitly.

A Centered two-point estimates

This appendix proves the three rows used in Section 3.5. We retain the explicit product estimate at the endpoint as an intermediate step, then divide by a controlled entropy denominator.

A.1 The estimates to be proved

For $0 < t < 1$, integration of $-\eta'(u) = \operatorname{atanh} u$ and midpoint Jensen give

$$K_0 = t \int_0^1 \operatorname{atanh}(vt) dv \geq pt \ln \frac{1 + (1-p)t}{1 - (1-p)t}. \quad (\text{A.1})$$

The quantitative interior estimate is

$$K_0 M_0 - h(p)^2 t^2 \geq \frac{1999}{1250000} pt^2, \quad 0 < t \leq 22/25, \quad 0 < p \leq 1/20. \quad (\text{A.2})$$

Appendix A.2 proves this from (A.1), monotonicity in p , five endpoint comparisons, and $\ln 20 < 749/250$. The explicit surplus will give the final term in Theorem 2.4.

Near deterministic posteriors. For $22/25 \leq t < 1$, set

$$\varepsilon = \frac{1-t}{2}, \quad r = \frac{\varepsilon}{p}. \quad (\text{A.3})$$

The centered posteriors are $\varepsilon, 1 - \varepsilon$, so $J_0 = h(\varepsilon)$ and $M_0 = h(p + \rho\varepsilon)$. Keeping the actual J_0 matters: the entropy present before mixing helps control the quadratic.

When $0 < \varepsilon \leq p$, Appendix A.4 gives

$$q_0(d) \geq \frac{23}{3600} pt, \quad 0 \leq d \leq \frac{1}{2}. \quad (\text{A.4})$$

Here there is also a useful gain-dominance relation:

$$K_0 \geq \frac{J_0}{2}, \quad \text{and therefore} \quad K_0 \geq \frac{M_0}{3}. \quad (\text{A.5})$$

To verify it, put $L = \ln(1/p)$. Since $\varepsilon \leq p \leq 1/20$, we have $t \geq 9/10$, $J_0 \leq h(p) \leq p(L+1)$, and the endpoint gain estimate (A.12) gives

$$K_0 \geq pt[L - \ln 3 + 3/5] \geq \frac{9}{10} p(L - 1/2).$$

For $L \geq 299/100$,

$$\frac{9}{10}(L - 1/2) - \frac{L+1}{2} = \frac{2}{5}L - \frac{19}{20} \geq \frac{123}{500} > 0.$$

This proves (A.5). Since $1/3 > 1/4$, the second part of Lemma 3.2 applies with $D = 1/2$.

When $p \leq \varepsilon \leq 3/50$, Appendix A.5 instead proves the product margin

$$K_0 M_0 - h(p)^2 t^2 \geq \frac{453}{5000} p^2 t^2. \quad (\text{A.6})$$

The proof first moves to the worst boundary by monotonicity. Two affine lower bounds then control its positive product. The final calculation is (A.19); no parameter grid is used.

We use the entropy decomposition

$$h(s) = s[\ln(1/s) + \mathcal{B}(s)], \quad \mathcal{B}(s) = -\frac{1-s}{s} \ln(1-s), \quad \mathcal{B}(0) = 1. \quad (\text{A.7})$$

Expansion of $-\ln(1-s)$ gives

$$\mathcal{B}(s) = 1 - \sum_{j \geq 1} \frac{s^j}{j(j+1)}. \quad (\text{A.8})$$

In particular,

$$1 - s \leq \mathcal{B}(s) \leq 1 - s/2 \leq 1, \quad \mathcal{B}(s) \geq 1 - s/2 - s^2/4 \quad (0 \leq s \leq 1/3). \quad (\text{A.9})$$

For the last bound, the tail starting at $j = 2$ is at most $s^2/[6(1-s)] \leq s^2/4$. The lower bound $1 - s$ follows from $-\ln(1-s) \geq s$. These formulas will also be used in Appendix D.

A.2 Interior separations

Here we can use a product margin. The proof first bounds the mixing gain by its midpoint derivative, then shows that the resulting lower bound is worst at the largest permitted noise. What remains is a one-variable estimate, covered by five monotonicity intervals.

Let $0 < t \leq 22/25$ and $0 < p \leq 1/20$. Define

$$c(t) = \ln \frac{1 + (19/20)t}{1 - (19/20)t}, \quad L = \ln(1/p).$$

The gain bound (A.1) gives $K_0 \geq pt c(t)$. Since $M_0 = J_0 + K_0$,

$$\frac{K_0 M_0}{pt^2} \geq \frac{c(t)}{t} \eta(t) + p c(t)^2. \quad (\text{A.10})$$

Also $h(p) \leq p(L + 1)$. Consequently,

$$\frac{K_0 M_0 - h(p)^2 t^2}{pt^2} \geq \frac{c(t)}{t} \eta(t) - p[(L + 1)^2 - c(t)^2].$$

For fixed t , the derivative of $p[(L + 1)^2 - c(t)^2]$ with respect to p is

$$L^2 - 1 - c(t)^2 > 0,$$

because $L \geq \ln 20 > 299/100$ and $c(t) \leq c(22/25) < 5/2$. The lower bound is therefore worst at $p = 1/20$. We first verify

$$\frac{c(t)}{t} \eta(t) + \frac{c(t)^2}{20} \geq \frac{4}{5}. \quad (\text{A.11})$$

The ratio $c(t)/t$ increases (expand it as a power series with nonnegative coefficients), while $\eta(t)$ decreases. The following bounds therefore hold on entire intervals:

Interval for t	Lower bound for $c(t)/t$	For $\eta(t)$	For $c(t)$
$(0, 1/2]$	$19/10$	$11/20$	0
$[1/2, 3/4]$	2	$3/8$	1
$[3/4, 4/5]$	$7/3$	$8/25$	$7/4$
$[4/5, 21/25]$	$39/16$	$11/40$	$39/20$
$[21/25, 22/25]$	$325/126$	$9/40$	$13/6$

For each row, multiply the second and third entries and add one twentieth of the square of the fourth. The result is at least $4/5$; in the row giving equality, the underlying bounds are strict. The endpoint logarithms are

$$\begin{aligned} c(1/2) &= \ln(59/21) > 1, & c(3/4) &= \ln(137/23) > 7/4, \\ c(4/5) &= \ln(22/3) > 39/20, & c(21/25) &= \ln(899/101) > 13/6, \\ c(22/25) &= \ln(459/41) < 5/2. \end{aligned}$$

Appendix E verifies these and the five entropy values. This proves (A.11). The sharper finite bound $\ln 20 < 749/250$ now gives

$$\frac{K_0 M_0 - h(p)^2 t^2}{pt^2} \geq \frac{4}{5} - \frac{(\ln 20 + 1)^2}{20} \geq \frac{4}{5} - \frac{(999/250)^2}{20} = \frac{1999}{1250000}.$$

This is (A.2). The table is an endpoint argument justified by monotonicity, not a sample of signs on a continuous domain.

A.3 Endpoint separations: a common lower bound

As the centered posteriors approach 0 and 1, logarithms rather than quadratic approximations control the entropy. The ratio $r = \varepsilon/p$ separates two situations: mixing supplies the dominant entropy when $r \leq 1$, whereas enough pre-existing entropy remains to control a discriminant when $r \geq 1$. The bounds below are designed for those two uses.

Let $22/25 \leq t < 1$, and set

$$\varepsilon = \frac{1-t}{2}, \quad r = \frac{\varepsilon}{p}, \quad u_e = \ln(1/\varepsilon), \quad L = \ln(1/p) = u_e + \ln r.$$

Here $0 < \varepsilon \leq 3/50$, $u_e > 14/5$, and $L > 299/100$. The subscript on u_e distinguishes this endpoint logarithm from the curvature coordinate used in Section 3.2.

In (A.1), the numerator is at least $459/250$, whose logarithm exceeds $3/5$. The denominator satisfies

$$1 - (1-p)t = p + 2(1-p)\varepsilon \leq p(1+2r).$$

Furthermore, (A.9) gives $\mathcal{B}(\varepsilon) \geq 9691/10000 > 19/20$. Consequently,

$$\frac{K_0}{pt} \geq \alpha := L - \ln(1+2r) + \frac{3}{5}, \quad \frac{J_0}{pt} = \frac{r[u_e + \mathcal{B}(\varepsilon)]}{t} \geq r\beta, \quad \beta := u_e + \frac{19}{20}. \quad (\text{A.12})$$

The last inequality uses $0 < t \leq 1$ and $u_e + \mathcal{B}(\varepsilon) > 0$: replacing $1/t$ by one preserves a lower bound. Since $M_0 = J_0 + K_0$ and $h(p)/p \leq L + 1$, for $d \geq 0$ the normalized centered gap obeys

$$\frac{q_0(d)}{pt} \geq \mathcal{Q}(d) := \alpha + (\alpha + r\beta)d^2 - 2(L+1)d. \quad (\text{A.13})$$

We next control this one quadratic in the two scale regimes.

A.4 When $\varepsilon \leq p$: use the endpoint of the allowed interval

Suppose $0 < r \leq 1$. Here $\alpha > 0$, because $L > 299/100$ and $\ln(1+2r) \leq \ln 3 < 11/10$. Also $\beta > 0$.

At fixed p , and therefore fixed L , we have $u_e = L - \ln r$. Differentiating gives

$$\frac{\partial}{\partial r} [\alpha + r\beta - 2(L+1)] = u_e - \frac{1}{20} - \frac{2}{1+2r} > 0.$$

At $r = 1$, the expression in square brackets is $-\ln 3 - 9/20 < 0$. Thus

$$\mathcal{Q}'(1/2) = \alpha + r\beta - 2(L+1) < 0.$$

Because the quadratic coefficient is positive, $\mathcal{Q}$ decreases throughout $[0, 1/2]$. Its minimum on that interval is therefore

$$\mathcal{Q}(1/2) = \frac{1+r}{4}L - \frac{5}{4}\ln(1+2r) - \frac{r}{4}\ln r + \frac{19r}{80} - \frac{1}{4}. \quad (\text{A.14})$$

For $r \leq 1/3$, use $\ln(1+2r) \leq 2r$, $-\ln r \geq \ln 3 > 1$, and $L \geq 299/100$. They give

$$\mathcal{Q}(1/2) \geq \frac{199}{400} - \frac{253r}{200} \geq \frac{91}{1200} > 0.$$

For $1/3 \leq r \leq 1$, use the tangent bound

$$\ln(1+2r) \leq \ln 3 + \frac{2}{3}(r-1) < \frac{11}{10} + \frac{2}{3}(r-1)$$

and discard $-r \ln r \geq 0$. Then

$$\mathcal{Q}(1/2) \geq \frac{91r}{600} - \frac{53}{1200} \geq \frac{23}{3600} > 0.$$

Together with (A.13), these bounds prove (A.4).

A.5 When $\varepsilon \geq p$: use the discriminant

Suppose $r \geq 1$. The estimate

$$\ln(2 + 1/r) = \ln 3 + \ln \left(1 - \frac{r-1}{3r}\right) \leq \frac{11}{10} - \frac{r-1}{3r}$$

gives

$$\alpha \geq a_*(u_e, r) := u_e - \frac{1}{2} + \frac{r-1}{3r} > 0. \quad (\text{A.15})$$

Both coefficients in (A.12) are positive. To prove (A.6), it therefore suffices to show

$$\mathcal{R}(u_e, r) := a_*(u_e, r)[a_*(u_e, r) + r(u_e + 19/20)] - (u_e + \ln r + 1)^2 \geq \frac{453}{5000}. \quad (\text{A.16})$$

Indeed, (A.12) then implies

$$\frac{K_0 M_0 - h(p)^2 t^2}{p^2 t^2} \geq \alpha(\alpha + r\beta) - (L+1)^2 \geq \mathcal{R}(u_e, r) \geq \frac{453}{5000}.$$

We first reduce (A.16) to a boundary. At fixed u_e , $\partial_r a_* \geq 0$, and direct differentiation gives

$$\begin{aligned} \partial_r \mathcal{R} &= (\partial_r a_*)[2a_* + r(u_e + 19/20)] + a_*(u_e + 19/20) - \frac{2(u_e + \ln r + 1)}{r} \\ &\geq (u_e - 1/2)(u_e + 19/20) - 2(u_e + 1) \\ &= u_e^2 - \frac{31}{20}u_e - \frac{99}{40} \geq \frac{41}{40} > 0. \end{aligned}$$

In the second line, we used that $(u_e + \ln r + 1)/r$ decreases for $r \geq 1$. The final quadratic increases for $u_e \geq 14/5$.

Put $c = 299/100$. The actual domain satisfies $u_e \geq 14/5$, $r \geq 1$, and $u_e + \ln r \geq c$. We prove this uniform lower bound on the enlarged domain, so its lower boundary may lie outside the original noise constraint; this relaxation is harmless for a lower bound. The boundary in r is

$$r = \max(1, e^{c-u_e}).$$

If $u_e \geq c$, monotonicity reduces to $r = 1$. There,

$$\mathcal{R}(u_e, 1) = u_e^2 - \frac{51}{20}u_e - \frac{49}{40} \geq \frac{453}{5000} > 0.$$

For the remaining boundary, write $v = c - u_e \in [0, 19/100] \subset [0, 1/5]$ and $r = e^v$. The negative term in (A.16) is then exactly $(c+1)^2$. We bound only its positive product, keeping that negative term fixed.

Since $e^v \geq 1 + v$, the two positive factors have the affine lower bounds

$$\begin{aligned} a_*(c-v, e^v) &\geq c-v - \frac{1}{2} + \frac{v}{3(1+v)} \geq \frac{249}{100} - \frac{13}{18}v, \\ e^v(c-v+19/20) &\geq (1+v) \left(\frac{197}{50} - v \right) \geq \frac{197}{50} + \frac{137}{50}v. \end{aligned} \quad (\text{A.17})$$

The first line uses $1+v \leq 6/5$; the second uses $v^2 \leq v/5$. Both lower bounds are positive. Adding them to bound the second factor of the product in (A.16), we obtain

$$\begin{aligned} \mathcal{R}(c-v, e^v) &\geq \left(\frac{249}{100} - \frac{13}{18}v \right) \left(\frac{643}{100} + \frac{454}{225}v \right) - \left(\frac{399}{100} \right)^2 \\ &= \frac{453}{5000} + \frac{17117}{45000}v - \frac{2951}{2025}v^2. \end{aligned} \quad (\text{A.18})$$

One more use of $v^2 \leq v/5$ gives the short sign certificate

$$\mathcal{R}(c - v, e^v) \geq \frac{453}{5000} + \frac{36013}{405000}v > 0. \quad (\text{A.19})$$

This proves (A.16) everywhere in the enlarged domain and completes (A.6). The endpoint argument has used monotonicity, two affine comparisons, and a quadratic expansion. It does not require a polynomial root count or a continuous-domain numerical certificate.

A.6 The normalized endpoint bound

To obtain an order- p surplus, the product estimate just proved is divided by the actual centered entropy, rather than by a constant upper bound on that entropy. We prove

$$K_0 - \ell^2/M_0 \geq (453/40000)pt \quad (p \leq \varepsilon \leq 3/50).$$

Let $p \leq \varepsilon \leq 3/50$, and set

$$u = \ln(1/\varepsilon), \quad r = \varepsilon/p, \quad c = 299/100.$$

Here u is the endpoint logarithm, not the midpoint displacement ν . The domain satisfies

$$u \geq 14/5, \quad r \geq 1, \quad u + \ln r \geq c.$$

Define

$$a_* = u - \frac{1}{2} + \frac{r-1}{3r}, \quad \beta = u + \frac{19}{20}, \quad B_* = a_* + r\beta, \quad z_* = u + \ln r + 1, \quad \Phi = a_* - \frac{z_*^2}{B_*}. \quad (\text{A.20})$$

All denominators and positive factors are positive. The endpoint estimates (A.12)–(A.15) imply

$$K_0 \geq pt a_*, \quad M_0 \geq pt B_*, \quad h(p)/p \leq z_*.$$

Hence $K_0 - \ell^2/M_0 \geq pt \Phi$. It suffices to prove

$$\Phi(u, r) \geq \frac{453}{40000} > \frac{1}{500} \quad (\text{A.21})$$

on the enlarged domain above. This also ensures the nonnegative numerator required in (3.20).

At fixed u , direct differentiation gives

$$\partial_r \Phi = \partial_r a_* + \frac{z_*}{rB_*^2} [z_* r \partial_r B_* - 2B_*].$$

Since $\partial_r a_* = 1/(3r^2) \geq 0$, $a_* \leq u - 1/6$, and $z_* - 2 \geq u - 1$, the bracket satisfies

$$\begin{aligned} z_* r \partial_r B_* - 2B_* &= z_* r \partial_r a_* + r\beta(z_* - 2) - 2a_* \\ &\geq (u + 19/20)(u - 1) - 2(u - 1/6) \\ &= u^2 - \frac{41}{20}u - \frac{37}{60} \geq \frac{89}{60} > 0. \end{aligned}$$

The final polynomial increases for $u \geq 14/5$. Thus Φ increases with r , and its minimum at fixed u is at $r = \max(1, e^{c-u})$.

If $u \geq c$, take $r = 1$. Then

$$\Phi(u, 1) = \frac{u^2 - 51u/20 - 49/40}{2u + 9/20}.$$

Its derivative has positive numerator $2u^2 + 9u/10 + 521/400$. It follows that

$$\Phi(u, 1) \geq \Phi(c, 1) = \frac{453}{32150} > \frac{453}{40000}.$$

If $14/5 \leq u \leq c$, write $v = c - u \in [0, 19/100]$ and $r = e^v$. The function $\mathcal{R}$ in (A.16) is exactly $a_* B_* - z_*^2$, and the proof of (A.16) establishes $\mathcal{R} \geq 453/5000$ on this enlarged domain. Moreover,

$$a_* < 3, \quad \beta < 4, \quad r = e^v \leq \frac{1}{1-v} \leq \frac{5}{4},$$

where the exponential bound follows term by term from its power series for $0 \leq v < 1$. Therefore $B_* < 8$, and

$$\Phi = \frac{\mathcal{R}}{B_*} \geq \frac{453}{40000}.$$

This proves (A.21). Equation (3.20) gives $S_{p,d}(a, b) \geq (453/40000)pt \geq pt^2/500$.

B The large-coordinate comparison

We supply the two facts used after (4.5): concavity in the output bias and the endpoint comparison (4.6).

B.1 Centering the information upper bound

The quantity to center is the explicit upper bound $U_\rho(m, b)$ on information, not the Boolean function and not its mutual information. At fixed coordinate coefficient b , an even concave function of m is largest at $m = 0$. The following derivative calculation proves that concavity on the full feasible bias interval.

The function $U_\rho(m, b)$ is even in m . In the interior of $|m| + b \leq 1$, differentiation yields

$$\partial_m^2 U_\rho(m, b) = \frac{-\rho^2 D_0}{(1 - m^2)[1 - (m + \rho b)^2][1 - (m - \rho b)^2]}, \quad (\text{B.1})$$

where

$$D_0 = (1 - m^2)^2 - b^2[1 + 3m^2 + \rho^2(1 - m^2)] + \rho^2 b^4.$$

By evenness it suffices to take $m \geq 0$. Set $z = (1 - m)^2 - b^2 \geq 0$. Expanding gives the nonnegative decomposition

$$\begin{aligned} D_0 &= 2m(1 - m)^3(1 - \rho^2) \\ &\quad + z[(1 - \rho^2)(1 + 3m^2) + 4\rho^2 m] + \rho^2 z^2 \geq 0. \end{aligned} \quad (\text{B.2})$$

The denominator in (B.1) is positive in the interior. Thus U_ρ is concave in m , and evenness gives $U_\rho(m, b) \leq U_\rho(0, b)$. Boundary cases follow by continuity.

B.2 Reducing the coefficient range to its endpoints

Concavity in the coordinate coefficient reduces the comparison to $b = 13/20$ and $b = 1$. At the nontrivial endpoint, divide by $\rho^2(1 - \rho^2)$ and expand in ρ^2 . The coefficients from degree two onward have one sign, so a second concavity argument reduces the correlation interval to its endpoints.

Put

$$G_\rho(b) = (1 - \rho^2)\eta(\rho b) - (1 - b\rho^2)\eta(\rho).$$

For fixed ρ , this is concave in b and $G_\rho(1) = 0$. It suffices to prove $G_\rho(13/20) \geq 0$ throughout $0 \leq \rho \leq 9/10$; the chord then covers all $b \in [13/20, 1]$.

Let $x = \rho^2$ and $a_n = 1/[2n(2n - 1)]$. Expanding (1.7) and dividing by $x(1 - x)$ gives

$$\frac{G_\rho(b)}{x(1 - x)} = \sum_{k \geq 0} [a_{k+1}(1 - b^{2k+2}) - (1 - b)T_k] x^k, \quad T_k = \sum_{n \geq k+1} a_n. \quad (\text{B.3})$$

For clarity, the coefficient of x^k after division by $1 - x$ is

$$a_{k+1}(1 - b^{2k+2}) + (1 - b) \sum_{j=1}^k a_j - (1 - b) \ln 2.$$

The identity $\sum_{j \geq 1} a_j = \ln 2$ turns it into the displayed coefficient. Here $n = k + 1$ is only a change of summation index, not a restriction on input dimension. A geometric-series integral gives

$$T_k = \int_0^1 \frac{v^{2n-2}}{1 + v} dv \geq \frac{1}{2(2n - 1)} = na_n.$$

At $b = 13/20$, every coefficient of x^k for $k \geq 2$ in (B.3) is negative: $(1 - b)(k + 1) \geq 21/20 > 1 > 1 - b^{2k+2}$. The function defined by the series is therefore concave on $[0, 81/100]$. Termwise differentiation is justified inside the radius of convergence.

Its value at $x = 0$ is

$$(1 - b) \left(\frac{1 + b}{2} - \ln 2 \right) > \frac{7}{20} \left(\frac{33}{40} - \frac{7}{10} \right) = \frac{7}{160} > 0.$$

At the other endpoint, the unnormalized numerator is

$$\begin{aligned} G_{9/10}(13/20) &= \frac{19}{100} h(83/400) - \frac{947}{2000} h(1/20) \\ &> \frac{19}{200} - \frac{947}{10000} = \frac{3}{10000} > 0. \end{aligned}$$

Here $83/400 > 1/5$ and both are below $1/2$, so monotonicity gives $h(83/400) > h(1/5) > 1/2$. The bounds $h(1/5) > 1/2$ and $h(1/20) < 1/5$ are verified in Appendix E. Concavity of the normalized expression places it above the chord between its positive endpoints. This proves (4.6) for $\rho > 0$. At $\rho = 0$, it holds with equality.

C A polynomial bound for capped sign sums

This appendix proves Lemma 4.5. The method is an explicit polynomial majorant and exact moments; it imposes no bound on the number of signs.

C.1 Global positivity by a factorization

We need an upper envelope of $|x|$ whose expectation can be bounded from finitely many moments. Evenness treats negative x automatically. The squared factors in the remainder below impose contact at three positive points; the remaining factor has positive coefficients. This certifies the envelope on the whole real line before any expectation is taken.

Define the even polynomial

$$P(x) = \frac{4336x^{10} - 117880x^8 + 1187295x^6 - 5572600x^4 + 15433465x^2 + 2903076}{12862500}. \quad (\text{C.1})$$

For $x \geq 0$, direct expansion gives

$$P(x) - x = \frac{(x-3)^2(x-2)^2(2x-1)^2}{12862500} \cdot (1084x^4 + 11924x^3 + 50475x^2 + 99674x + 80641) \geq 0. \quad (\text{C.2})$$

All coefficients of the quartic are positive. Evenness then proves $P(x) \geq |x|$ for every real x . There is no root count and no sampled positivity test. The contact points $1/2, 2, 3$ explain the squared factors: P agrees with $|x|$ and its derivative at those positive points.

C.2 Moments and one convex quadratic

Let $R = \sum_j a_j X_j$, $\sum_j a_j^2 = 1$, and $|a_j| \leq 2/5$. Put

$$t = \sum_j a_j^4, \quad u = \sum_j a_j^6, \quad v = \sum_j a_j^8, \quad w = \sum_j a_j^{10}, \quad c = \frac{4}{25}.$$

These are local moment variables, unrelated to posterior separation in Section 3. The coefficient cap gives

$$0 \leq t \leq c, \quad 0 \leq u \leq ct, \quad 0 \leq v \leq cu, \quad 0 \leq w \leq cv.$$

The even moments $M_j = \mathbb{E}R^j$ are

$$\begin{aligned} M_0 &= M_2 = 1, \\ M_4 &= 3 - 2t, \\ M_6 &= 15 - 30t + 16u, \\ M_8 &= 105 - 420t + 140t^2 + 448u - 272v, \\ M_{10} &= 945 - 6300t + 6300t^2 + 10080u - 6720tu - 12240v + 7936w. \end{aligned}$$

For completeness, expand

$$\log \mathbb{E}e^{sR} = \sum_j \log \cosh(a_j s) = \frac{s^2}{2} - \frac{ts^4}{12} + \frac{us^6}{45} - \frac{17vs^8}{2520} + \frac{31ws^{10}}{14175} + O(s^{12}).$$

Differentiating $\mathbb{E}e^{sR} = \exp(\log \mathbb{E}e^{sR})$ gives the recurrence $M_k = \sum_{j=1}^k \binom{k-1}{j-1} \kappa_j M_{k-j}$, where the even cumulants are $1, -2t, 16u, -272v, 7936w$ and the odd ones vanish. This yields the displayed moments in every finite dimension.

Substitution in (C.1) gives the exact expression

$$\begin{aligned} \mathbb{E}P(R) &= \frac{1}{6431250} (5406800t^2 - 14568960tu - 1140425t \\ &\quad + 4946680u - 10504640v + 17205248w + 5574143). \end{aligned} \quad (\text{C.3})$$

The coefficient of w is positive. Using $w \leq cv$, the two last moment terms satisfy

$$-10504640v + 17205248w \leq -\frac{193795008}{25}v \leq 0.$$

For $0 \leq t \leq c$, the coefficient of u is at least $4946680 - 14568960c = 13078232/5 > 0$. We may therefore set $u = ct$ for an upper bound. Consequently,

$$\mathbb{E}|R| \leq \mathbb{E}P(R) \leq Q_*(t) := \frac{15378832t^2 - 1744781t + 27870715}{32156250}. \quad (\text{C.4})$$

This quadratic is convex. On $[0, c]$ it is at most the larger endpoint value, and

$$\frac{7}{8} - Q_*(0) = \frac{212803}{25725000} > 0, \quad \frac{7}{8} - Q_*(c) = \frac{126225509}{26796875000} > 0.$$

Thus $\mathbb{E}|R| \leq 7/8$, proving Lemma 4.5. Relaxing the moment constraints only enlarged an upper bound; no claim that an extremizing moment tuple is realizable is needed.

D Entropy comparison on the correlation interval

We prove Lemma 5.3. The common device is to compare entropy with $\lambda\psi_\rho$ using the representation (A.7). It reduces a statement for all posterior values to a scalar inequality in the coefficient λ .

D.1 The strict anchor at $\rho = 3/5$

Put $\lambda_0 = 39/40$ and

$$D(t) = \eta(t) - \lambda_0\psi_{3/5}(t).$$

For $t \geq 1/3$, write $y = (1 - t)/2 \leq 1/3$. From (A.8),

$$\mathcal{B}(y) \geq 1 - y/2 - \frac{y^2}{6(1 - y)} \geq 1 - \frac{7y}{12}.$$

Since $\psi_\rho(t) = 2y(2 - \rho^2 - 2y)$, for $y > 0$ this yields

$$\frac{D(t)}{y} \geq -\ln y + 1 + (4\lambda_0 - 7/12)y - 2\lambda_0(2 - 9/25).$$

For every $c > 0$, the minimum of $-\ln y + cy$ over $y > 0$ is $1 + \ln c$. Therefore the right side is at least

$$\ln(199/60) - 599/500 > 0. \quad (\text{D.1})$$

The certified rational margin is recorded in Appendix E.

For $0 \leq t \leq 1/3$,

$$D''(t) = \frac{39}{20} - \frac{1}{1 - t^2} > 0, \quad D'(1/3) = \frac{299}{1000} - \frac{\ln 2}{2} < 0.$$

Hence D' is negative on the whole interval, and the minimum of D there is

$$D(1/3) = \ln 3 - \frac{2}{3} \ln 2 - \frac{949}{1500} > 0. \quad (\text{D.2})$$

Appendix E verifies the finite signs in (D.1)–(D.2). At $t = 1$, both sides of (5.11) vanish. This proves the anchor comparison.

D.2 A coefficient criterion for the remaining interval

Rather than minimize the ratio $\eta(t)/\psi_\rho(t)$ separately for every ρ , we derive a sufficient condition on a coefficient λ . The substitution $y = (1 - t)/2$ extracts the endpoint logarithm. Minimizing the elementary expression $-\ln y + cy$ then removes the posterior variable entirely.

Now take $3/5 \leq \rho \leq 9/10$. The function $\mathcal{B}$ is concave by (A.8). Its chord between 0 and $1/2$ gives

$$\mathcal{B}(y) \geq 1 - 2(1 - \ln 2)y \geq 1 - \frac{5}{8}y, \quad 0 \leq y \leq 1/2,$$

since $\ln 2 > 11/16$.

For $\lambda > 5/32$ and $y = (1 - t)/2 > 0$, it follows that

$$\begin{aligned} \eta(t) - \lambda\psi_\rho(t) &\geq y[-\ln y + 1 + (4\lambda - 5/8)y - 2\lambda(2 - \rho^2)] \\ &\geq y[\ln(4\lambda - 5/8) + 2 - 2\lambda(2 - \rho^2)]. \end{aligned}$$

Thus a sufficient criterion for $\eta(t) \geq \lambda\psi_\rho(t)$ for every t is

$$\ln(4\lambda - 5/8) + 2 - 2\lambda(2 - \rho^2) \geq 0. \quad (\text{D.3})$$

The case $y = 0$ follows by continuity.

The coefficient required by (5.10) is

$$\lambda_{\text{req}}(\rho) = \frac{\eta(\rho)}{(1 - \rho)\Lambda_\rho}.$$

Using $\mathcal{B}(y) \leq 1 - y/2$ at $y = (1 - \rho)/2$, we obtain the upper bound

$$\lambda_{\text{req}}(\rho) \leq \bar{\lambda}(\rho) := \frac{\ln(2/(1 - \rho)) + 3/4 + \rho/4}{2\Lambda_\rho}. \quad (\text{D.4})$$

We will prove (D.3) for a still larger, explicitly constructed coefficient. Since $\psi_\rho \geq 0$, that proves a stronger entropy lower bound and therefore the required one. This direction of comparison is essential.

D.3 Why four coefficients cover the whole interval

The required coefficient is bounded above by a convex function of ρ . Chords between four rational upper bounds therefore suffice. Concavity of the logarithm gives a lower chord in the opposite direction. The remaining sign is a cubic polynomial on each interval, certified by its positive Bernstein coefficients. Thus the table below represents complete-interval estimates, not samples.

We first verify that $\bar{\lambda}$ is convex. Let N be the numerator in (D.4). Differentiating gives

$$\bar{\lambda}'' = \frac{N''\Lambda^2 - 2N'\Lambda\Lambda' + 2N(\Lambda')^2 - N\Lambda\Lambda''}{2\Lambda^3},$$

with all functions evaluated at ρ . Here N, N', N'', Λ are positive and $\Lambda'' < 0$. If $\Lambda' \leq 0$, every term in the numerator is nonnegative and the first is positive. If $\Lambda' > 0$, then $\rho < 20/31$ and

$$\Lambda' \leq \frac{7}{100}, \quad N' < \frac{13}{4}, \quad N'' \geq \frac{25}{4}, \quad \Lambda > \frac{5}{4}.$$

These imply $N''\Lambda - 2N'\Lambda' > 0$, and the remaining terms are nonnegative. Thus $\bar{\lambda}'' > 0$ throughout the interval.

Choose four knots ρ_j , upper coefficients λ_j , and lower logarithm bounds s_j :

j	ρ_j	λ_j	s_j
0	3/5	19/20	11553/10000
1	3/4	287/250	689/500
2	17/20	1377/1000	15857/10000
3	9/10	1561/1000	17261/10000

The finite comparisons, verified in Appendix E, are

$$\lambda_j > \bar{\lambda}(\rho_j), \quad s_j < \ln(4\lambda_j - 5/8). \quad (\text{D.5})$$

Appendix E records certified margins for the narrowest of these comparisons. Between consecutive knots, write

$$\rho = (1 - z)\rho_j + z\rho_{j+1}, \quad \lambda = (1 - z)\lambda_j + z\lambda_{j+1}, \quad 0 \leq z \leq 1.$$

Convexity of $\bar{\lambda}$ places it below its chord, so $\lambda \geq \bar{\lambda}(\rho)$. Concavity of the logarithm places it above its chord, so the left side of (D.3) is at least

$$P_j(z) = (1 - z)s_j + zs_{j+1} + 2 - 2\lambda(2 - \rho^2). \quad (\text{D.6})$$

Each P_j is a cubic. Direct expansion gives

$$P_j(z) = c_{j0}(1 - z)^3 + 3c_{j1}z(1 - z)^2 + 3c_{j2}z^2(1 - z) + c_{j3}z^3,$$

with the following positive coefficients:

j	c_{j0}	c_{j1}	c_{j2}	c_{j3}
0	393/10000	829/75000	1249/60000	31/400
1	31/400	1683/40000	11161/300000	13493/200000
2	13493/200000	21353/600000	493/30000	273/25000

The four basis functions are nonnegative and sum to one on $[0, 1]$. Hence $P_j(z) > 0$ throughout that interval. This is an algebraic certificate covering the complete interval, not an interpolation of sampled signs.

We have proved (D.3) for $\lambda \geq \bar{\lambda}(\rho) \geq \lambda_{\text{req}}(\rho)$. Since $\psi_\rho(t) \geq 0$, the resulting bound $\eta(t) \geq \lambda\psi_\rho(t)$ implies (5.10). This completes Lemma 5.3.

E Rational bounds for logarithms and constants

The finite comparisons in this manuscript use rational arithmetic and a convergent logarithm series with an explicit remainder. This appendix states the procedure so that no rounded decimal is a proof premise.

For rational $q \geq 1$, write $q = 2^k v$ with $1 \leq v < 2$, set $z = (v - 1)/(v + 1)$, and define

$$L_N(v) = 2 \sum_{j=0}^{N-1} \frac{z^{2j+1}}{2j+1},$$

$$U_N(v) = L_N(v) + \frac{2z^{2N+1}}{(2N+1)(1-z^2)}.$$

The positive series for $\ln((1+z)/(1-z))$ gives

$$L_N(v) \leq \ln v \leq U_N(v). \quad (\text{E.1})$$

For the powers-of-two reduction, evaluate the same series at $v = 2$, where $z = 1/3$; the displayed remainder is valid there as well. In particular set

$$\ell_2 = L_{12}(2), \quad u_2 = U_{12}(2).$$

For $q = 2^k v \geq 1$ with $1 \leq v < 2$, the actual working enclosures are

$$k\ell_2 + L_{12}(v) \leq \ln q \leq ku_2 + U_{12}(v).$$

The coarse claims $56/81 < \ln 2 < 7/10$ in the table below are *outputs to verify*, not the enclosures used in these multiples. For an argument below one, reverse the bounds using $\ln q = -\ln(1/q)$. Every operation is on rational numbers.

With $N = 12$, these bounds prove all of the following comparisons:

Argument q	Lower bound for $\ln q$	Upper bound for $\ln q$
2	56/81	7/10
3	13/12	11/10
20	299/100	749/250
50/3	14/5	—
459/250	3/5	—
59/21	1	—
137/23	7/4	—
22/3	39/20	—
899/101	13/6	—
459/41	—	5/2
199/60	599/500	—
127/40	11553/10000	—
3967/1000	689/500	—
4883/1000	15857/10000	—
5619/1000	17261/10000	—

The four upper bounds for $\bar{\lambda}(\rho_j)$ are equivalently

$$\ln \frac{2}{1-\rho_j} < 2\Lambda_{\rho_j} \lambda_j - \frac{3}{4} - \frac{\rho_j}{4}.$$

Their logarithm arguments are 5, 8, 40/3, 20, so (E.1) verifies them by the same procedure.

For rational $a = r/s \in (0, 1)$, entropy can be evaluated through

$$s h(r/s) = s \ln s - r \ln r - (s - r) \ln(s - r).$$

The same rational enclosures prove

$$\begin{aligned} h(1/4) &> 11/20, & h(1/8) &> 3/8, & h(1/10) &> 8/25, \\ h(2/25) &> 11/40, & h(3/50) &> 9/40, \\ h(1/5) &> 1/2, & h(1/20) &< 1/5, \end{aligned}$$

as well as

$$\ln 3 - \frac{2}{3} \ln 2 > \frac{949}{1500}, \quad \ln 2 - \frac{39}{40} \frac{2}{5} \frac{1321}{1000} < \frac{9}{50}.$$

The quantitative scalar margin uses the rational comparisons

$$\frac{1999}{1250000} > \frac{7}{5000}, \quad \frac{453}{5000} > \frac{7}{250}, \quad \frac{23}{3600} > \frac{1}{500}.$$

These comparisons also imply the weaker $p^2/25$ surplus. The stronger coefficient $p/500$ in Theorem 2.4 follows from the retained-denominator argument and the comparisons in its proof. The coefficient is deliberately conservative.

Margins for the closest comparisons

The following lower bounds are strict rational inequalities verified by the preceding $N = 12$ procedure. They are not rounded decimal estimates. For each row, the computed enclosure of the difference has width less than 10^{-11} .

Positive difference	Certified lower bound
$\ln(127/40) - 11553/10000$	$7/10^6$
$\ln(3967/1000) - 689/500$	$1/100000$
$\lambda_1 - \bar{\lambda}(3/4)$	$1/20000$
$\lambda_2 - \bar{\lambda}(17/20)$	$1/40000$
$749/250 - \ln 20$	$1/4000$
$h(1/5) - 1/2$	$1/2500$
$\ln(199/60) - 599/500$	$9/10000$

These margins document the precision required at (D.5), (D.1), and (A.11). They do not change any constant used in the proof. The finite comparison in Section 6, $U_0 < 179/1000$, uses the same enclosure of $\ln 2$.

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